

EXAMPLE 3 | **BB** Antigenic cartography (continued) Given the coordinate system overlaid on the two-dimensional antigenic space in Figure 14, find the components of the vector represented by the directed line segment from the cluster A in 1987 to cluster B in 1989.

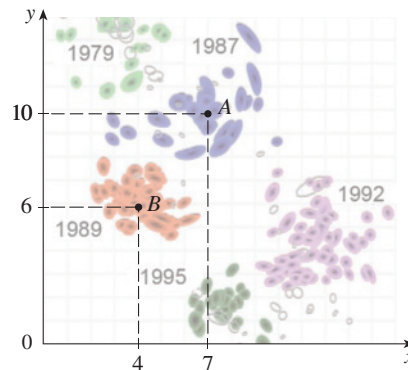


FIGURE 14

SOLUTION By result (1), the vector corresponding to $\overrightarrow{AB}$ is

$$\mathbf{a} = [4 - 7, 6 - 10] = [-3, -4]$$

The **magnitude** or **length** of a vector $\mathbf{a}$ is the length of any of its representations and is denoted by the symbol $|\mathbf{a}|$ or $\|\mathbf{a}\|$. By using the distance formula to compute its length, we obtain the following formulas.

The length of the two-dimensional vector $\mathbf{a} = [a_1, a_2]$ is

$$|\mathbf{a}| = \sqrt{a_1^2 + a_2^2}$$

The length of the three-dimensional vector $\mathbf{a} = [a_1, a_2, a_3]$ is

$$|\mathbf{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

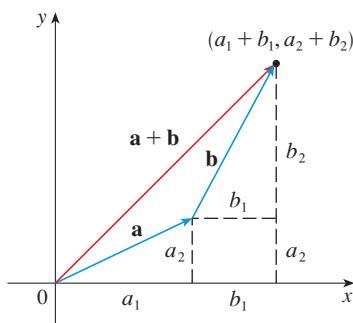


FIGURE 15

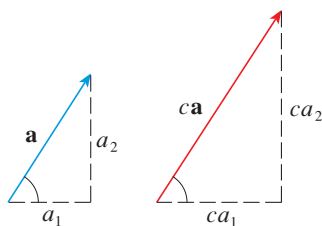


FIGURE 16

How do we add vectors algebraically? Figure 15 shows that if $\mathbf{a} = [a_1, a_2]$ and $\mathbf{b} = [b_1, b_2]$, then the sum is $\mathbf{a} + \mathbf{b} = [a_1 + b_1, a_2 + b_2]$, at least for the case where the components are positive. In other words, *to add algebraic vectors we add their components*. Similarly, *to subtract vectors we subtract components*. From the similar triangles in Figure 16 we see that the components of $c\mathbf{a}$ are ca_1 and ca_2 . So *to multiply a vector by a scalar we multiply each component by that scalar*.

If $\mathbf{a} = [a_1, a_2]$ and $\mathbf{b} = [b_1, b_2]$, then

$$\mathbf{a} + \mathbf{b} = [a_1 + b_1, a_2 + b_2] \quad \mathbf{a} - \mathbf{b} = [a_1 - b_1, a_2 - b_2]$$

$$c\mathbf{a} = [ca_1, ca_2]$$

Similarly, for three-dimensional vectors,

$$[a_1, a_2, a_3] + [b_1, b_2, b_3] = [a_1 + b_1, a_2 + b_2, a_3 + b_3]$$

$$[a_1, a_2, a_3] - [b_1, b_2, b_3] = [a_1 - b_1, a_2 - b_2, a_3 - b_3]$$

$$c[a_1, a_2, a_3] = [ca_1, ca_2, ca_3]$$

A common alternative notation for vectors in three dimensions makes use of unit vectors in the x -, y -, and z -directions. By convention these three special vectors are denoted by $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$, respectively. Exercises 8.2.30–32 show how any three-dimensional vector can be written in terms of $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$.

A **unit vector** is a vector whose length is 1. In general, if $\mathbf{a} \neq \mathbf{0}$, then the unit vector that has the same direction as $\mathbf{a}$ is

$$(2) \quad \mathbf{u} = \frac{1}{|\mathbf{a}|} \mathbf{a} = \frac{\mathbf{a}}{|\mathbf{a}|}$$

EXAMPLE 4 | Find the unit vector in the direction of the vector $\mathbf{a} = [2, -1, -2]$.

SOLUTION The given vector has length

$$|\mathbf{a}| = \sqrt{2^2 + (-1)^2 + (-2)^2} = \sqrt{9} = 3$$

so, by Equation 2, the unit vector with the same direction is

$$\frac{1}{3}[2, -1, -2] = \left[\frac{2}{3}, -\frac{1}{3}, -\frac{2}{3}\right]$$

EXAMPLE 5 | If $\mathbf{a} = [4, 0, 3]$ and $\mathbf{b} = [-2, 1, 5]$, find $|\mathbf{a}|$ and the vectors $\mathbf{a} + \mathbf{b}$, $\mathbf{a} - \mathbf{b}$, $3\mathbf{b}$, and $2\mathbf{a} + 5\mathbf{b}$.

SOLUTION $|\mathbf{a}| = \sqrt{4^2 + 0^2 + 3^2} = \sqrt{25} = 5$

$$\mathbf{a} + \mathbf{b} = [4, 0, 3] + [-2, 1, 5]$$

$$= [4 + (-2), 0 + 1, 3 + 5] = [2, 1, 8]$$

$$\mathbf{a} - \mathbf{b} = [4, 0, 3] - [-2, 1, 5]$$

$$= [4 - (-2), 0 - 1, 3 - 5] = [6, -1, -2]$$

$$3\mathbf{b} = 3[-2, 1, 5] = [3(-2), 3(1), 3(5)] = [-6, 3, 15]$$

$$2\mathbf{a} + 5\mathbf{b} = 2[4, 0, 3] + 5[-2, 1, 5]$$

$$= [8, 0, 6] + [-10, 5, 25] = [-2, 5, 31]$$

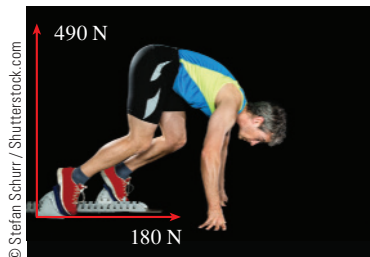


FIGURE 17

EXAMPLE 6 | **Biomechanics** A force is represented by a vector because it has a magnitude (in newtons) and a direction. If more than one force is acting on an object, the **resultant force** is the vector sum of these forces. Consider the horizontal and vertical forces exerted by an athlete at the start of the 100-meter sprint as shown in Figure 17.

- What is the resultant force vector exerted by the athlete?
- What is the total magnitude of the force exerted by the athlete?

SOLUTION

(a) The force vector in the horizontal direction is $[180, 0]$ and in the vertical direction, $[0, 490]$. The resultant force vector is therefore $[180, 0] + [0, 490] = [180, 490]$.

(b) The total magnitude of the force is the length of the resultant force vector. This is $\sqrt{180^2 + 490^2} \approx 522$ newtons.

We denote by V_2 the set of all two-dimensional vectors and by V_3 the set of all three-dimensional vectors. More generally, V_n is the set of all n -dimensional vectors where an n -dimensional vector is an ordered n -tuple:

$$\mathbf{a} = [a_1, a_2, \dots, a_n]$$

Vectors in n dimensions are used to list various quantities in an organized way. For instance, the components of a six-dimensional vector

$$\mathbf{p} = [p_1, p_2, p_3, p_4, p_5, p_6]$$

might represent the expression levels of six different genes.

and $a_1, a_2, \dots, a_n$ are real numbers called the components of $\mathbf{a}$. Addition and scalar multiplication are defined in terms of components just as for the cases $n = 2$ and $n = 3$. Likewise, the length of a vector from V_n is calculated by using the distance formula on page 493.

Properties of Vectors If $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ are vectors in V_n and c and d are scalars, then

$$1. \mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$$

$$2. \mathbf{a} + (\mathbf{b} + \mathbf{c}) = (\mathbf{a} + \mathbf{b}) + \mathbf{c}$$

$$3. \mathbf{a} + \mathbf{0} = \mathbf{a}$$

$$4. \mathbf{a} + (-\mathbf{a}) = \mathbf{0}$$

$$5. c(\mathbf{a} + \mathbf{b}) = c\mathbf{a} + c\mathbf{b}$$

$$6. (c + d)\mathbf{a} = c\mathbf{a} + d\mathbf{a}$$

$$7. (cd)\mathbf{a} = c(d\mathbf{a})$$

$$8. 1\mathbf{a} = \mathbf{a}$$

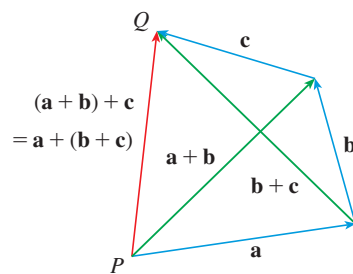


FIGURE 18

These eight properties of vectors can be readily verified either geometrically or algebraically. For instance, Property 1 can be seen from Figure 4 (it's equivalent to the Parallelogram Law) or as follows for the case $n = 2$:

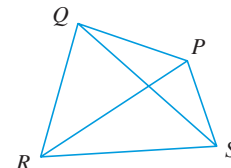
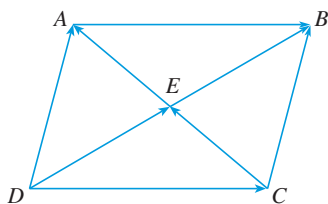
$$\begin{aligned} \mathbf{a} + \mathbf{b} &= [a_1, a_2] + [b_1, b_2] = [a_1 + b_1, a_2 + b_2] \\ &= [b_1 + a_1, b_2 + a_2] = [b_1, b_2] + [a_1, a_2] \\ &= \mathbf{b} + \mathbf{a} \end{aligned}$$

We can see why Property 2 (the associative law) is true by looking at Figure 18 and applying the Triangle Law several times: The vector $\overrightarrow{PQ}$ is obtained either by first constructing $\mathbf{a} + \mathbf{b}$ and then adding $\mathbf{c}$ or by adding $\mathbf{a}$ to the vector $\mathbf{b} + \mathbf{c}$.

EXERCISES 8.2

Note: Vector notations using square brackets $[1, 1]$ and angle brackets $\langle 1, 1 \rangle$ are equivalent. You may see either notation in your online homework.

- Are the following quantities vectors or scalars? Explain.
 - The cost of a theater ticket
 - The current in a river
 - The initial flight path from Houston to Dallas
 - The population of the world
- What is the relationship between the point $(4, 7)$ and the vector $[4, 7]$? Illustrate with a sketch.
- Name all the equal vectors in the parallelogram shown.
- Write each combination of vectors as a single vector.
 - $\overrightarrow{PQ} + \overrightarrow{QR}$
 - $\overrightarrow{RP} + \overrightarrow{PS}$
 - $\overrightarrow{QS} - \overrightarrow{PS}$
 - $\overrightarrow{RS} + \overrightarrow{SP} + \overrightarrow{PQ}$



- Copy the vectors in the figure and use them to draw the following vectors.

- $\mathbf{u} + \mathbf{v}$
- $\mathbf{v} + \mathbf{w}$

- $\mathbf{u} - \mathbf{v}$
- $\mathbf{w} + \mathbf{v} + \mathbf{u}$



6. Copy the vectors in the figure and use them to draw the following vectors.

- (a) $\mathbf{a} + \mathbf{b}$ (b) $\mathbf{a} - \mathbf{b}$
 (c) $\frac{1}{2}\mathbf{a}$ (d) $-3\mathbf{b}$
 (e) $\mathbf{a} + 2\mathbf{b}$ (f) $2\mathbf{b} - \mathbf{a}$



- 7–10 Find a vector $\mathbf{a}$ with representation given by the directed line segment $\overrightarrow{AB}$. Draw $\overrightarrow{AB}$ and the equivalent representation starting at the origin.

7. $A(-1, 3)$, $B(2, 2)$ 8. $A(2, 1)$, $B(0, 6)$
 9. $A(0, 3, 1)$, $B(2, 3, -1)$ 10. $A(4, 0, -2)$, $B(4, 2, 1)$

- 11–14 Find the sum of the given vectors and illustrate geometrically.

11. $[-1, 4]$, $[6, -2]$ 12. $[-2, -1]$, $[5, 7]$
 13. $[0, 1, 2]$, $[0, 0, -3]$ 14. $[-1, 0, 2]$, $[0, 4, 0]$

- 15–18 Find $\mathbf{a} + \mathbf{b}$, $2\mathbf{a} + 3\mathbf{b}$, $|\mathbf{a}|$, and $|\mathbf{a} - \mathbf{b}|$.

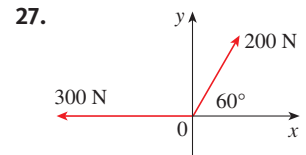
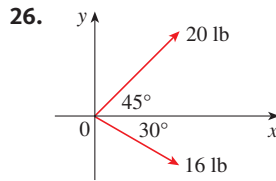
15. $\mathbf{a} = [5, -12]$, $\mathbf{b} = [-3, -6]$
 16. $\mathbf{a} = [4, 1]$, $\mathbf{b} = [1, -2]$
 17. $\mathbf{a} = [1, 2, -3]$, $\mathbf{b} = [-2, -1, 5]$
 18. $\mathbf{a} = [2, -4, 4]$, $\mathbf{b} = [0, 2, -1]$

- 19–21 Find a unit vector that has the same direction as the given vector.

19. $[-3, 7]$ 20. $[-4, 2, 4]$
 21. $[8, -1, 4]$

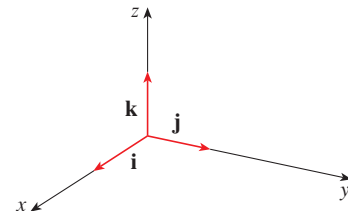
22. Find a vector that has the same direction as $[-2, 4, 2]$ but has length 6.
 23. If $\mathbf{v}$ lies in the first quadrant and makes an angle $\pi/3$ with the positive x -axis and $|\mathbf{v}| = 4$, find $\mathbf{v}$ in component form.
 24. If a child pulls a sled through the snow on a level path with a force of 50 N exerted at an angle of 38° above the horizontal, find the horizontal and vertical components of the force.
 25. A quarterback throws a football with angle of elevation 40° and speed 60 ft/s. Find the horizontal and vertical components of the velocity vector.

- 26–27 Find the magnitude of the resultant force and the angle it makes with the positive x -axis.



28. The magnitude of a velocity vector is called *speed*. Suppose that a wind is blowing from the direction N45°W at a speed of 50 km/h. (This means that the direction from which the wind blows is 45° west of the northerly direction.) A pilot is steering a plane in the direction N60°E at an airspeed (speed in still air) of 250 km/h. The *true course*, or *track*, of the plane is the direction of the resultant of the velocity vectors of the plane and the wind. The *ground speed* of the plane is the magnitude of the resultant. Find the true course and the ground speed of the plane.
 29. A woman walks due west on the deck of a ship at 3 mi/h. The ship is moving north at a speed of 22 mi/h. Find the speed and direction of the woman relative to the surface of the water.

- 30–32 The unit vectors in V_3 that coincide with the coordinate axes are called the **standard basis vectors** and are denoted by $\mathbf{i} = [1, 0, 0]$, $\mathbf{j} = [0, 1, 0]$, and $\mathbf{k} = [0, 0, 1]$. (See the figure.)



If $\mathbf{a} = [a_1, a_2, a_3]$, we can write

$$\begin{aligned}\mathbf{a} &= [a_1, 0, 0] + [0, a_2, 0] + [0, 0, a_3] \\ &= a_1[1, 0, 0] + a_2[0, 1, 0] + a_3[0, 0, 1]\end{aligned}$$

Therefore any vector $\mathbf{a}$ in V_3 can be expressed in terms of the standard basis vectors as $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$.

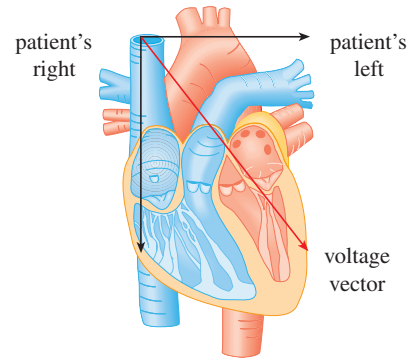
30. Express the following vectors in terms of the standard basis vectors.
 (a) $[-1, 4]$ (b) $[5, 7]$
 (c) $[-2, 1, 2]$ (d) $[-1, 0, 2]$

31. If $\mathbf{a} = \mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$ and $\mathbf{b} = 4\mathbf{i} + 7\mathbf{k}$, evaluate the following in terms of the standard basis vectors.
- (a) $\mathbf{a} + \mathbf{b}$ (b) $\mathbf{a} - \mathbf{b}$
 (c) $2\mathbf{a} + 3\mathbf{b}$ (d) $5\mathbf{a} - 7\mathbf{b}$
32. Find the unit vector that points in the same direction as the given vector and express it in terms of the standard basis vectors.
- (a) $\mathbf{i} + \mathbf{j}$ (b) $\mathbf{i} + \mathbf{j} + \mathbf{k}$
 (c) $2\mathbf{i} - \mathbf{k}$ (d) $4\mathbf{i} + 6\mathbf{j} - \mathbf{k}$
33. Find the unit vectors that are parallel to the tangent line to the parabola $y = x^2$ at the point $(2, 4)$.
34. (a) Find the unit vectors that are parallel to the tangent line to the curve $y = 2 \sin x$ at the point $(\pi/6, 1)$.
 (b) Find the unit vectors that are perpendicular to the tangent line.
 (c) Sketch the curve $y = 2 \sin x$ and the vectors in parts (a) and (b), all starting at $(\pi/6, 1)$.
35. (a) Draw the vectors $\mathbf{a} = [3, 2]$, $\mathbf{b} = [2, -1]$, and $\mathbf{c} = [7, 1]$.
 (b) Show, by means of a sketch, that there are scalars s and t such that $\mathbf{c} = s\mathbf{a} + t\mathbf{b}$.
 (c) Use the sketch to estimate the values of s and t .
 (d) Find the exact values of s and t .
36. Suppose that $\mathbf{a}$ and $\mathbf{b}$ are nonzero vectors that are not parallel and $\mathbf{c}$ is any vector in the plane determined by $\mathbf{a}$ and $\mathbf{b}$. Give a geometric argument to show that $\mathbf{c}$ can be written as $\mathbf{c} = s\mathbf{a} + t\mathbf{b}$ for suitable scalars s and t . Then give an argument using components.
37. Suppose $\mathbf{a}$ is a three-dimensional unit vector in the first octant that starts at the origin and makes angles of 60° and 72° with the positive x - and y -axes, respectively. Express $\mathbf{a}$ in terms of its components.
38. **Biomechanics** Two sprinters of equal mass leave the starting blocks with the following horizontal and vertical force vectors:

Runner	Horizontal	Vertical
Runner 1	150 N	300 N
Runner 2	200 N	250 N

Newton's Second Law states that force is equal to mass times acceleration ($\mathbf{F} = m\mathbf{a}$). Which runner has the greater acceleration out of the blocks?

39. **Vectorcardiography** As the heart beats it generates differences in electrical potential (that is, voltage) across the body. Cardiologists view the voltage at any point in time during a heartbeat as a vector. During ventricular contraction in healthy individuals this vector points downward and to the left of the patient (see the figure). Abnormalities in either the magnitude or direction can be used to diagnose cardiac problems.



We can think of the voltage vector as being the resultant vector of a vertical and a horizontal component of voltage (see the figure). Conductive abnormalities in different parts of the heart can reduce the voltage in either of these directions, thereby altering the direction and magnitude of the resultant voltage vector.

- (a) A *left anterior hemiblock* reverses the direction of the vertical component of voltage and reduces the magnitude of the horizontal component. Describe how such a condition could be diagnosed based on the resultant voltage vector.
- (b) A *left posterior hemiblock* reverses the direction of the horizontal component of voltage. Describe how such a condition could be diagnosed based on the resultant voltage vector.
40. Suppose a vector $\mathbf{a}$ makes angles α , β , and γ with the positive x -, y -, and z -axes, respectively. Find the components of $\mathbf{a}$ and show that $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$. (The numbers $\cos \alpha$, $\cos \beta$, and $\cos \gamma$ are called the *direction cosines* of $\mathbf{a}$.)
41. If $\mathbf{r} = [x, y, z]$ and $\mathbf{r}_0 = [x_0, y_0, z_0]$, describe the set of all points (x, y, z) such that $|\mathbf{r} - \mathbf{r}_0| = 1$.
42. If $\mathbf{r} = [x, y]$, $\mathbf{r}_1 = [x_1, y_1]$, and $\mathbf{r}_2 = [x_2, y_2]$, describe the set of all points (x, y) such that $|\mathbf{r} - \mathbf{r}_1| + |\mathbf{r} - \mathbf{r}_2| = k$, where $k > |\mathbf{r}_1 - \mathbf{r}_2|$.
43. Figure 18 gives a geometric demonstration of Property 2 of vectors. Use components to give an algebraic proof of this fact for the case $n = 2$.
44. Prove Property 5 of vectors algebraically for the case $n = 3$. Then use similar triangles to give a geometric proof.
45. Use vectors to prove that the line joining the midpoints of two sides of a triangle is parallel to the third side and half its length.
46. **Antigenic cartography** The Triangle Inequality for vectors (see Exercise 8.3.48) is

$$|\mathbf{a} + \mathbf{b}| \leq |\mathbf{a}| + |\mathbf{b}|$$

Suppose that $\mathbf{a}$ denotes the vector of antigenic change in influenza from year 2012 to 2013, and $\mathbf{b}$ denotes the vector of antigenic change from 2013 to 2014. Explain what the Triangle Inequality means in terms of this antigenic change.

8.3 The Dot Product

So far we have added two vectors and multiplied a vector by a scalar. The question arises: Is it possible to multiply two vectors so that their product is a useful quantity? One such product is the dot product.

(1) Definition If $\mathbf{a} = [a_1, a_2, a_3]$ and $\mathbf{b} = [b_1, b_2, b_3]$, then the **dot product** of $\mathbf{a}$ and $\mathbf{b}$ is the number $\mathbf{a} \cdot \mathbf{b}$ given by

$$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3$$

Thus, to find the dot product of $\mathbf{a}$ and $\mathbf{b}$, we multiply corresponding components and add. The result is not a vector. It is a real number, that is, a scalar. For this reason, the dot product is sometimes called the **scalar product**. Although Definition 1 is given for three-dimensional vectors, the dot product of two-dimensional vectors is defined in a similar fashion:

$$[a_1, a_2] \cdot [b_1, b_2] = a_1b_1 + a_2b_2$$

Likewise, for n -dimensional vectors we have:

$$[a_1, \dots, a_n] \cdot [b_1, \dots, b_n] = a_1b_1 + \dots + a_nb_n$$

EXAMPLE 1

$$[2, 4] \cdot [3, -1] = 2(3) + 4(-1) = 2$$

$$[-1, 7, 4] \cdot \left[6, 2, -\frac{1}{2}\right] = (-1)(6) + 7(2) + 4\left(-\frac{1}{2}\right) = 6$$

$$[1, 2, -3] \cdot [0, 2, -1] = 1(0) + 2(2) + (-3)(-1) = 7$$

The dot product obeys many of the laws that hold for ordinary products of real numbers.

(2) Properties of the Dot Product If $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ are vectors in V_3 and c is a scalar, then

- | | |
|---|---|
| 1. $\mathbf{a} \cdot \mathbf{a} = \mathbf{a} ^2$ | 2. $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$ |
| 3. $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$ | 4. $(c\mathbf{a}) \cdot \mathbf{b} = c(\mathbf{a} \cdot \mathbf{b}) = \mathbf{a} \cdot (c\mathbf{b})$ |
| 5. $\mathbf{0} \cdot \mathbf{a} = 0$ | |

These properties are easily proved using Definition 1. For instance, here are the proofs of Properties 1 and 3:

$$1. \mathbf{a} \cdot \mathbf{a} = a_1^2 + a_2^2 + a_3^2 = |\mathbf{a}|^2$$

$$\begin{aligned} 3. \mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) &= [a_1, a_2, a_3] \cdot [b_1 + c_1, b_2 + c_2, b_3 + c_3] \\ &= a_1(b_1 + c_1) + a_2(b_2 + c_2) + a_3(b_3 + c_3) \\ &= a_1b_1 + a_1c_1 + a_2b_2 + a_2c_2 + a_3b_3 + a_3c_3 \\ &= (a_1b_1 + a_2b_2 + a_3b_3) + (a_1c_1 + a_2c_2 + a_3c_3) \\ &= \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c} \end{aligned}$$

The proofs of the remaining properties are left as exercises.

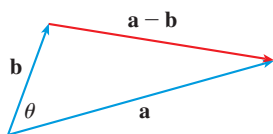


FIGURE 1

The dot product $\mathbf{a} \cdot \mathbf{b}$ can be given a geometric interpretation in terms of the angle θ between $\mathbf{a}$ and $\mathbf{b}$. This angle is defined to be the angle between the representations of $\mathbf{a}$ and $\mathbf{b}$ that start at the origin, where $0 \leq \theta \leq \pi$ (see Figure 1). From the Law of Cosines for the two vectors we have:

$$(3) \quad |\mathbf{a} - \mathbf{b}|^2 = |\mathbf{a}|^2 + |\mathbf{b}|^2 - 2|\mathbf{a}||\mathbf{b}|\cos\theta$$

Using Properties 1, 2, and 3 of the dot product, we can rewrite the left side of this equation as follows:

$$\begin{aligned} |\mathbf{a} - \mathbf{b}|^2 &= (\mathbf{a} - \mathbf{b}) \cdot (\mathbf{a} - \mathbf{b}) \\ &= \mathbf{a} \cdot \mathbf{a} - \mathbf{a} \cdot \mathbf{b} - \mathbf{b} \cdot \mathbf{a} + \mathbf{b} \cdot \mathbf{b} \\ &= |\mathbf{a}|^2 - 2\mathbf{a} \cdot \mathbf{b} + |\mathbf{b}|^2 \end{aligned}$$

Therefore Equation 3 gives

$$\begin{aligned} |\mathbf{a}|^2 - 2\mathbf{a} \cdot \mathbf{b} + |\mathbf{b}|^2 &= |\mathbf{a}|^2 + |\mathbf{b}|^2 - 2|\mathbf{a}||\mathbf{b}|\cos\theta \\ -2\mathbf{a} \cdot \mathbf{b} &= -2|\mathbf{a}||\mathbf{b}|\cos\theta \end{aligned}$$

or

(4) An Alternative Formula for the Dot Product

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos\theta$$

where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$ ($0 \leq \theta \leq \pi$). (θ is the smaller angle between the two vectors when drawn from the same initial point.)

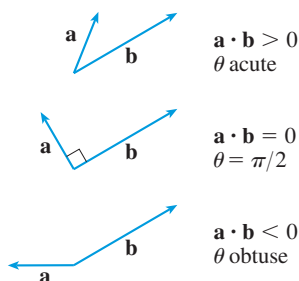


FIGURE 2

TEC Visual 8.3A shows an animation of Figure 2.

Notice that, if the dot product of two nonzero vectors is zero, then $\cos\theta = 0$. Thus $\theta = \pi/2$ and the two vectors are perpendicular (sometimes called **orthogonal**; see Figure 2). Also, because $\cos\theta > 0$ if $0 \leq \theta < \pi/2$ and $\cos\theta < 0$ if $\pi/2 < \theta \leq \pi$, we see that $\mathbf{a} \cdot \mathbf{b}$ is positive for $\theta < \pi/2$ and negative for $\theta > \pi/2$. We can therefore think of $\mathbf{a} \cdot \mathbf{b}$ as measuring the extent to which $\mathbf{a}$ and $\mathbf{b}$ point in the same direction. The dot product $\mathbf{a} \cdot \mathbf{b}$ is positive if $\mathbf{a}$ and $\mathbf{b}$ point in the same general direction, 0 if they are perpendicular, and negative if they point in generally opposite directions (see Figure 2). In the extreme case where $\mathbf{a}$ and $\mathbf{b}$ point in exactly the same direction we have $\theta = 0$, so $\cos\theta = 1$ and

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}|$$

If $\mathbf{a}$ and $\mathbf{b}$ point in exactly opposite directions, then $\theta = \pi$ and so $\cos\theta = -1$ and $\mathbf{a} \cdot \mathbf{b} = -|\mathbf{a}||\mathbf{b}|$.

EXAMPLE 2 | Show that $[2, 2, -1]$ is perpendicular to $[5, -4, 2]$.

SOLUTION Since

$$[2, 2, -1] \cdot [5, -4, 2] = 2(5) + 2(-4) + (-1)(2) = 0$$

these vectors are perpendicular. ■

EXAMPLE 3 | Find the angle between the vectors $\mathbf{a} = [2, 2, -1]$ and $\mathbf{b} = [5, -3, 2]$.

SOLUTION Let θ be the required angle. Since

$$|\mathbf{a}| = \sqrt{2^2 + 2^2 + (-1)^2} = 3 \quad \text{and} \quad |\mathbf{b}| = \sqrt{5^2 + (-3)^2 + 2^2} = \sqrt{38}$$

and since

$$\mathbf{a} \cdot \mathbf{b} = 2(5) + 2(-3) + (-1)(2) = 2$$

we obtain

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|} = \frac{2}{3\sqrt{38}}$$

So the angle between $\mathbf{a}$ and $\mathbf{b}$ is

$$\theta = \cos^{-1}\left(\frac{2}{3\sqrt{38}}\right) \approx 1.46 \quad (\text{or } 84^\circ)$$

We now have techniques for comparing vectors both in terms of their magnitudes (in Section 8.2) and in terms of their directions (the dot product). Both can provide important information in a variety of biological contexts.

EXAMPLE 4 | **BB** Using the dot product for biological discovery¹

Recall from Example 8.1.2 that a genome expression profile gives the level of expression of each of a collection of genes. Thus, it can be represented as an n -dimensional vector, where n is the number of genes assayed. For some cells a large number of such expression profiles have been characterized in response to different, known biochemical compounds. These can then be used to discover the mode of action of new biochemical compounds by quantifying the genome expression profile induced by the new compound and determining which known profile it most closely resembles. The new compound likely affects cell function through a mechanism similar to that of the best-matching, known biochemical compound.

Consider expression profiles $\mathbf{a}$ and $\mathbf{b}$ that are induced by two known biochemical compounds. Suppose that four genes are assayed and the expression profiles are given by the vectors $\mathbf{a} = [2, 5, 0, 1]$ and $\mathbf{b} = [1, 2, 4, 3]$, where the expression levels are dimensionless numbers. Suppose further that a new biochemical compound induces an expression profile given by the vector $\mathbf{n} = [1, 0, 5, 2]$.

- Which known expression profile vector points in a direction most similar to that of $\mathbf{n}$: profile $\mathbf{a}$ or profile $\mathbf{b}$?
- What is the angle between profile $\mathbf{n}$ and the profile identified in part (a)?

SOLUTION The magnitudes of the profiles are given by

$$|\mathbf{a}| = \sqrt{2^2 + 5^2 + 0^2 + 1^2} = \sqrt{30}$$

$$|\mathbf{b}| = \sqrt{1^2 + 2^2 + 4^2 + 3^2} = \sqrt{30}$$

$$|\mathbf{n}| = \sqrt{1^2 + 0^2 + 5^2 + 2^2} = \sqrt{30}$$

1. F. Kuruvilla et al., "Vector Algebra in the Analysis of Genome-Wide Expression Data," *Genome Biology* 3 (2002): research0011.1–11.

Because all vectors have the same magnitude, $\sqrt{30}$, the dot product of $\mathbf{n}$ with either $\mathbf{a}$ or $\mathbf{b}$ will have a maximum value of 30 if they point in the same direction and a minimum value of -30 if they point in opposite directions.

(a) Calculating the dot products gives

$$\mathbf{n} \cdot \mathbf{a} = [1, 0, 5, 2] \cdot [2, 5, 0, 1] = 4$$

$$\mathbf{n} \cdot \mathbf{b} = [1, 0, 5, 2] \cdot [1, 2, 4, 3] = 27$$

Therefore $\mathbf{n}$ points in a direction most similar to $\mathbf{b}$. We conclude that the new compound likely affects the cell through a mechanism similar to that of the known biochemical compound whose profile is $\mathbf{b}$.

(b) The angle between $\mathbf{n}$ and $\mathbf{b}$ is given by $\cos^{-1}(27/30) \approx 0.45$ (or 25.8°). ■

The dot product also enables us to write equations of planes. A plane is determined by a point $P_0(x_0, y_0, z_0)$ in the plane and a vector $\mathbf{n} = [a, b, c]$ that is orthogonal to the plane. This orthogonal vector $\mathbf{n}$ is called a **normal vector**. If $P(x, y, z)$ is any point in the plane, then the vector $[x - x_0, y - y_0, z - z_0]$ is perpendicular to $\mathbf{n}$ and so

$$[a, b, c] \cdot [x - x_0, y - y_0, z - z_0] = 0$$

This gives us the following equation for the plane.

(5) An equation of the plane that passes through the point $P_0(x_0, y_0, z_0)$ and is perpendicular to the vector $[a, b, c]$ is

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

EXAMPLE 5 | Find an equation of the plane through the point $(2, 4, -1)$ with normal vector $\mathbf{n} = [2, 3, 4]$.

SOLUTION Putting $a = 2$, $b = 3$, $c = 4$, $x_0 = 2$, $y_0 = 4$, and $z_0 = -1$ in Equation 5, we see that an equation of the plane is

$$2(x - 2) + 3(y - 4) + 4(z + 1) = 0$$

or

$$2x + 3y + 4z = 12$$
 ■



FIGURE 3

■ Projections

In addition to providing a means of comparing the direction of vectors, the dot product is useful for other analyses. Consider the force exerted by the foot of a runner as he pushes off the ground. The force propels him both forward and upward as shown in Figure 3. This force is a vector because it has both magnitude and direction. But what component of this force acts in the forward direction? Put another way, can we construct a horizontal vector that corresponds to the same forward force? From Figure 3 we can see that such a vector will have an x -component equal to the x -component of the original force vector.

More generally, Figure 4 shows representations $\vec{PQ}$ and $\vec{PR}$ of two arbitrary vectors $\mathbf{a}$ and $\mathbf{b}$ with the same initial point P . If S is the base of the perpendicular from R to the line containing $\vec{PQ}$, then the vector with representation $\vec{PS}$ is called the **vector projection** of $\mathbf{b}$ onto $\mathbf{a}$ and is denoted by $\text{proj}_{\mathbf{a}} \mathbf{b}$. (You can think of it as a shadow of $\mathbf{b}$).

The scalar projection of $\mathbf{b}$ onto $\mathbf{a}$ (also called the **component of $\mathbf{b}$ along $\mathbf{a}$**) is defined to be the signed magnitude of the vector projection, which is the number $|\mathbf{b}| \cos \theta$, where

FIGURE 4
Vector projections

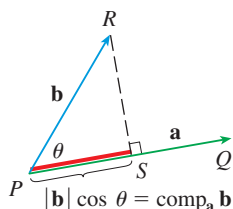
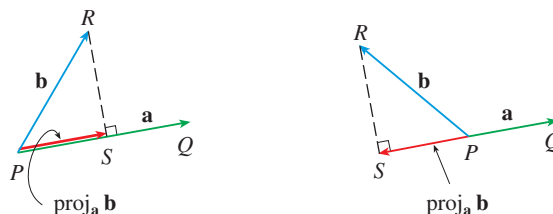


FIGURE 5
Scalar projection

θ is the angle between $\mathbf{a}$ and $\mathbf{b}$. (See Figure 5.) This is denoted by $\text{comp}_{\mathbf{a}} \mathbf{b}$. Observe that it is negative if $\pi/2 < \theta \leq \pi$.

The equation

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta = |\mathbf{a}| (|\mathbf{b}| \cos \theta)$$

shows that the dot product of $\mathbf{a}$ and $\mathbf{b}$ can be interpreted as the length of $\mathbf{a}$ times the scalar projection of $\mathbf{b}$ onto $\mathbf{a}$. Since

$$|\mathbf{b}| \cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}|} = \frac{\mathbf{a}}{|\mathbf{a}|} \cdot \mathbf{b}$$

the component of $\mathbf{b}$ along $\mathbf{a}$ can be computed by taking the dot product of $\mathbf{b}$ with the unit vector in the direction of $\mathbf{a}$. We summarize these ideas as follows.

$$\text{Scalar projection of } \mathbf{b} \text{ onto } \mathbf{a}: \quad \text{comp}_{\mathbf{a}} \mathbf{b} = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}|}$$

$$\text{Vector projection of } \mathbf{b} \text{ onto } \mathbf{a}: \quad \text{proj}_{\mathbf{a}} \mathbf{b} = \left(\frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}|} \right) \frac{\mathbf{a}}{|\mathbf{a}|} = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}|^2} \mathbf{a}$$

EXAMPLE 6 | Find the scalar projection and vector projection of $\mathbf{b} = [1, 1, 2]$ onto $\mathbf{a} = [-2, 3, 1]$.

SOLUTION Since $|\mathbf{a}| = \sqrt{(-2)^2 + 3^2 + 1^2} = \sqrt{14}$, the scalar projection of $\mathbf{b}$ onto $\mathbf{a}$ is

$$\text{comp}_{\mathbf{a}} \mathbf{b} = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}|} = \frac{(-2)(1) + 3(1) + 1(2)}{\sqrt{14}} = \frac{3}{\sqrt{14}}$$

The vector projection is this scalar projection times the unit vector in the direction of $\mathbf{a}$:

$$\text{proj}_{\mathbf{a}} \mathbf{b} = \frac{3}{\sqrt{14}} \frac{\mathbf{a}}{|\mathbf{a}|} = \frac{3}{14} \mathbf{a} = \left[-\frac{3}{7}, \frac{9}{14}, \frac{3}{14} \right]$$

EXAMPLE 7 | **BB** **Vectorcardiography and Einthoven's triangle** As the heart beats, it generates differences in electrical potential (that is, voltage) across the body. Cardiologists view the electrical potential generated by the heart at any point in time during a heartbeat as a vector. This vector points in the direction of the greatest potential, and its magnitude represents the voltage in this direction. Abnormalities in either the magnitude or direction can be used to diagnose cardiac problems.

In Chapter 1 we saw an example of measurements of the voltage between two points on a patient's body. Most electrocardiographs actually take readings from among at

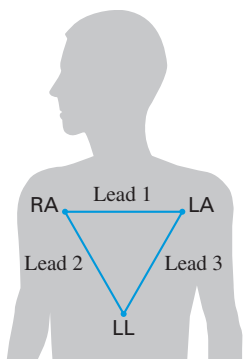


FIGURE 6
Einthoven's triangle

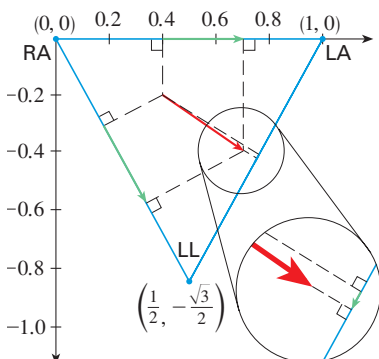


FIGURE 7
Red arrow is the heart voltage vector. Green arrows are the vector projections of the heart voltage vector onto the sides of Einthoven's triangle. The signed magnitude of each vector projection (that is, its scalar projection) is measured by the corresponding lead.

least three points on the body—the right arm (RA), the left arm (LA), and the left leg (LL). These points make up a triangle known as Einthoven's triangle, as shown in Figure 6. The voltage between all pairs of vertices of this triangle is measured continuously, with each side of the triangle being referred to as a *lead*. For example, the measured voltage between RA and LA at any point in time gives the voltage in the direction of a vector corresponding to the top edge of the triangle at that time (labeled lead 1 in Figure 6).

To capture this idea mathematically, Einthoven's triangle is typically viewed as an equilateral triangle with unit sides, and with vertices having the coordinates $(0, 0)$, $(1, 0)$, and $(1/2, -\sqrt{3}/2)$ as displayed in Figure 7. At any point in time, each of the three leads measures the voltage in the direction of the vector corresponding to that side of the triangle.

Now consider a heart voltage vector at some point during a heartbeat. We can't measure this vector directly, but we can use the measurements from the three leads to infer its magnitude and direction. In particular, the measurement from lead 1 represents the scalar projection of the heart voltage vector onto the edge of the triangle between RA and LA (see Figure 7). Likewise, the measurement from lead 2 represents the scalar projection of the heart voltage vector onto the edge of the triangle between RA and LL, and so on. Thus the voltages measured in the three leads at any time gives us information about the magnitude and direction of the heart voltage vector at that time.

Suppose that heart voltage vector at some time during the heartbeat points from location $(0.4, -0.2)$ to location $(0.7, -0.4)$ in the coordinate system containing Einthoven's triangle shown in Figure 7. Assume voltage is measured in mV.

- What is the voltage of the heart vector (that is, its magnitude) at this time?
- What will be the electrocardiograph reading for each of the three leads at this time?

SOLUTION

- From result 8.2.1, a representation of the heart voltage vector $\mathbf{h}$ is calculated as

$$\mathbf{h} = [0.7 - 0.4, -0.4 - (-0.2)] = [0.3, -0.2]$$

The magnitude of the heart vector is therefore

$$|\mathbf{h}| = \sqrt{0.3^2 + (-0.2)^2} \approx 0.36 \text{ mV}$$

- The three leads correspond to the three sides of Einthoven's triangle in Figure 7. From result 8.2.1, a vector representation of each side is

$$\mathbf{I}_1 = [1 - 0, 0 - 0] = [1, 0] \quad \text{for RA to LA}$$

$$\mathbf{I}_2 = \left[\frac{1}{2} - 0, -\frac{\sqrt{3}}{2} - 0 \right] = \left[\frac{1}{2}, -\frac{\sqrt{3}}{2} \right] \quad \text{for RA to LL}$$

$$\mathbf{I}_3 = \left[\frac{1}{2} - 1, -\frac{\sqrt{3}}{2} - 0 \right] = \left[-\frac{1}{2}, -\frac{\sqrt{3}}{2} \right] \quad \text{for LA to LL}$$

Notice that Einthoven's triangle is defined to have sides of unit length. We verify this as follows:

$$|\mathbf{I}_1| = \sqrt{1^2 + 0^2} = 1 \quad |\mathbf{I}_2| = \sqrt{\left(\frac{1}{2}\right)^2 + \left(-\frac{\sqrt{3}}{2}\right)^2} = 1$$

$$|\mathbf{I}_3| = \sqrt{\left(-\frac{1}{2}\right)^2 + \left(-\frac{\sqrt{3}}{2}\right)^2} = 1$$

The electrocardiograph reading from each lead is the scalar projection of the heart

voltage vector on that lead. Therefore the reading in lead 1 will be $\mathbf{l}_1 \cdot \mathbf{h}/|\mathbf{l}_1| = 0.3$. The reading in lead 2 will be $\mathbf{l}_2 \cdot \mathbf{h}/|\mathbf{l}_2| \approx 0.32$. Finally, the reading in lead 3 will be $\mathbf{l}_3 \cdot \mathbf{h}/|\mathbf{l}_3| \approx 0.02$. As expected from Figure 7, real readings from leads 1 and 2 are normally relatively large, while the reading from lead 3 is normally nearly zero because the heart voltage vector is usually almost orthogonal to this lead. ■

EXERCISES 8.3

Note: Vector notations using square brackets $[1, 1]$ and angle brackets $\langle 1, 1 \rangle$ are equivalent. You may see either notation in your online homework.

1. Which of the following expressions are meaningful? Which are meaningless? Explain.

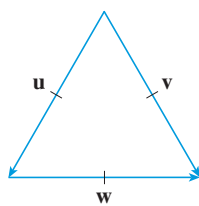
- (a) $(\mathbf{a} \cdot \mathbf{b}) \cdot \mathbf{c}$ (b) $(\mathbf{a} \cdot \mathbf{b})\mathbf{c}$
 (c) $|\mathbf{a}|(\mathbf{b} \cdot \mathbf{c})$ (d) $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c})$
 (e) $\mathbf{a} \cdot \mathbf{b} + \mathbf{c}$ (f) $|\mathbf{a}| \cdot (\mathbf{b} + \mathbf{c})$

2–10 Find $\mathbf{a} \cdot \mathbf{b}$. For Exercises 9–10, refer to the notation introduced in Exercises 8.2.30–32.

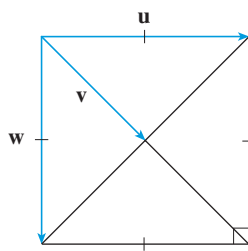
2. $|\mathbf{a}| = 3$, $|\mathbf{b}| = \sqrt{6}$, the angle between $\mathbf{a}$ and $\mathbf{b}$ is 45°
 3. $|\mathbf{a}| = 6$, $|\mathbf{b}| = 5$, the angle between $\mathbf{a}$ and $\mathbf{b}$ is $2\pi/3$
 4. $\mathbf{a} = [-2, 3]$, $\mathbf{b} = [0.7, 1.2]$
 5. $\mathbf{a} = [-2, \frac{1}{3}]$, $\mathbf{b} = [-5, 12]$
 6. $\mathbf{a} = [6, -2, 3]$, $\mathbf{b} = [2, 5, -1]$
 7. $\mathbf{a} = [4, 1, \frac{1}{4}]$, $\mathbf{b} = [6, -3, -8]$
 8. $\mathbf{a} = [p, -p, 2p]$, $\mathbf{b} = [2q, q, -q]$
 9. $\mathbf{a} = 2\mathbf{i} + \mathbf{j}$, $\mathbf{b} = \mathbf{i} - \mathbf{j} + \mathbf{k}$
 10. $\mathbf{a} = 3\mathbf{i} + 2\mathbf{j} - \mathbf{k}$, $\mathbf{b} = 4\mathbf{i} + 5\mathbf{k}$

11–12 If $\mathbf{u}$ is a unit vector, find $\mathbf{u} \cdot \mathbf{v}$ and $\mathbf{u} \cdot \mathbf{w}$.

11.



12.



13. Exercises 8.2.30–32 introduced the standard basis vectors $\mathbf{i} = [1, 0, 0]$, $\mathbf{j} = [0, 1, 0]$, and $\mathbf{k} = [0, 0, 1]$.

- (a) Show that $\mathbf{i} \cdot \mathbf{j} = \mathbf{j} \cdot \mathbf{k} = \mathbf{k} \cdot \mathbf{i} = 0$.
 (b) Show that $\mathbf{i} \cdot \mathbf{i} = \mathbf{j} \cdot \mathbf{j} = \mathbf{k} \cdot \mathbf{k} = 1$.

14. Population dynamics Suppose a population of fish contains a females that are 1 year old, b females between 2 and 4 years old, and c females more than 4 years old. During the summer each one-year-old female produces 15 offspring, each female between 2 and 4 years old produces 35 offspring, and each female more than 4 years old produces 20 offspring. If $\mathbf{a} = [a, b, c]$ and $\mathbf{p} = [15, 35, 20]$, what is the meaning of the dot product $\mathbf{a} \cdot \mathbf{p}$?

15–18 Find the angle between the vectors. (First find an exact expression and then approximate to the nearest degree.)

15. $\mathbf{a} = [-8, 6]$, $\mathbf{b} = [\sqrt{7}, 3]$

16. $\mathbf{a} = [\sqrt{3}, 1]$, $\mathbf{b} = [0, 5]$

17. $\mathbf{a} = [0, 1, 1]$, $\mathbf{b} = [1, 2, -3]$

18. $\mathbf{a} = [1, 2, -2]$, $\mathbf{b} = [4, 0, -3]$

19–20 Find, correct to the nearest degree, the three angles of the triangle with the given vertices.

19. $A(1, 0)$, $B(3, 6)$, $C(-1, 4)$

20. $D(0, 1, 1)$, $E(-2, 4, 3)$, $F(1, 2, -1)$

21–22 Determine whether the given vectors are orthogonal, parallel, or neither.

21. (a) $\mathbf{a} = [-5, 3, 7]$, $\mathbf{b} = [6, -8, 2]$

(b) $\mathbf{a} = [4, 6]$, $\mathbf{b} = [-3, 2]$

(c) $\mathbf{a} = [-1, 2, 5]$, $\mathbf{b} = [3, 4, -1]$

(d) $\mathbf{a} = [2, 6, -4]$, $\mathbf{b} = [-3, -9, 6]$

22. (a) $\mathbf{u} = [-3, 9, 6]$, $\mathbf{v} = [4, -12, -8]$

(b) $\mathbf{u} = [1, -1, 2]$, $\mathbf{v} = [2, -1, 1]$

(c) $\mathbf{u} = [a, b, c]$, $\mathbf{v} = [-b, a, 0]$

23. Use vectors to decide whether the triangle with vertices $P(1, -3, -2)$, $Q(2, 0, -4)$, and $R(6, -2, -5)$ is right-angled.

24. For what values of b are the vectors $[-6, b, 2]$ and $[b, b^2, b]$ orthogonal?

25. Find a unit vector that is orthogonal to both $[1, 1, 0]$ and $[1, 0, 1]$.

26. Find two unit vectors that make an angle of 60° with $\mathbf{v} = [3, 4]$.

27. Find an equation of the plane that passes through the origin and is perpendicular to the vector $[1, -2, 5]$.

28. Find an equation of the plane through the point $(-1, \frac{1}{2}, 3)$ with normal vector $[1, 4, 1]$.

29–32 Find the scalar and vector projections of $\mathbf{b}$ onto $\mathbf{a}$.

29. $\mathbf{a} = [3, -4]$, $\mathbf{b} = [5, 0]$

30. $\mathbf{a} = [1, 2]$, $\mathbf{b} = [-4, 1]$

31. $\mathbf{a} = [2, -1, 4]$, $\mathbf{b} = \left[0, 1, \frac{1}{2}\right]$

32. $\mathbf{a} = [1, 1, 1]$, $\mathbf{b} = [1, -1, 1]$

33. Show that the vector $\text{orth}_{\mathbf{a}}\mathbf{b} = \mathbf{b} - \text{proj}_{\mathbf{a}}\mathbf{b}$ is orthogonal to $\mathbf{a}$. (It is called an **orthogonal projection** of $\mathbf{b}$.)

34. For the vectors in Exercise 30, find $\text{orth}_{\mathbf{a}}\mathbf{b}$ and illustrate by drawing the vectors $\mathbf{a}$, $\mathbf{b}$, $\text{proj}_{\mathbf{a}}\mathbf{b}$, and $\text{orth}_{\mathbf{a}}\mathbf{b}$.

35. If $\mathbf{a} = [3, 0, -1]$, find a vector $\mathbf{b}$ such that $\text{comp}_{\mathbf{a}}\mathbf{b} = 2$.

36. Suppose that $\mathbf{a}$ and $\mathbf{b}$ are nonzero vectors.

- Under what circumstances is $\text{comp}_{\mathbf{a}}\mathbf{b} = \text{comp}_{\mathbf{b}}\mathbf{a}$?
- Under what circumstances is $\text{proj}_{\mathbf{a}}\mathbf{b} = \text{proj}_{\mathbf{b}}\mathbf{a}$?

37. Antigenic evolution Influenza viruses from North America and Asia are sampled in two successive years and plotted in two-dimensional antigenic space. The coordinates of each are (i) North American viruses: (2, 1) for 2013 and (4, 3) for 2014, and (ii) Asian viruses: (4, 18) for 2013 and (5, 17) for 2014.

- How does the magnitude of antigenic change compare between the two regions?
- How does the direction of antigenic change compare between the two regions?

38. Community ecology Community ecologists study the factors that determine the abundance of different species. The abundance of three algae species (in mg/mL) is quantified in two different lakes before and after an unusually hot summer. The coordinates of these four samples in $\mathbb{R}^3$ are (i) Lake A: (97, 84, 43) and (100, 80, 50) before and after summer, respectively, and (ii) Lake B: (23, 59, 22) and (20, 63, 15) before and after summer, respectively.

- How does the magnitude of the change in algal community compare between the two lakes?
- How does the direction of the change in algal community compare between the two lakes?

39. Genome expression profiles and drug design

Researchers are trying to design a drug that alters the expression level of three genes in a specific way. The desired change in expression, written as a vector in V_3 , is $[3, 9, -5]$, where all measurements are dimensionless. Suppose two potential drugs are being studied, and the expression profiles that they induce are (A) $[2, 4, 1]$ and (B) $[-2, 3, -5]$.

- What is the magnitude of the desired change in expression?
- Using scalar projections, determine the fraction of this desired change that is induced by each drug.
- Which drug is closest to having the desired effect?

40. Vectorcardiography In each of the following questions the heart voltage vector of a patient with different pathologies is given. Determine the electrocardiograph reading that would result in each of the three leads of Einthoven's triangle, and draw a sketch like that in Figure 7. Recall from Example 7 that vector representations of the sides of Einthoven's triangle are

$$\mathbf{l}_1 = [1, 0] \quad \mathbf{l}_2 = [1/2, -\sqrt{3}/2]$$

$$\mathbf{l}_3 = [-1/2, -\sqrt{3}/2]$$

- Left anterior hemiblock: $\mathbf{h} = [0.3, 0.2]$
- Left posterior hemiblock: $\mathbf{h} = [-0.3, -0.2]$
- Apical ischemia: $\mathbf{h} = [-0.3, 0.2]$
- Chronic obstructive pulmonary disease: $\mathbf{h} = [0.1, -0.0667]$

41. Find the angle between a diagonal of a cube and one of its edges.

42. Find the angle between a diagonal of a cube and a diagonal of one of its faces.

43. If $\mathbf{r} = [x, y, z]$, $\mathbf{a} = [a_1, a_2, a_3]$, and $\mathbf{b} = [b_1, b_2, b_3]$, show that the vector equation $(\mathbf{r} - \mathbf{a}) \cdot (\mathbf{r} - \mathbf{b}) = 0$ represents a sphere, and find its center and radius.

44. If $\mathbf{c} = |\mathbf{a}|\mathbf{b} + |\mathbf{b}|\mathbf{a}$, where $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ are all nonzero vectors, show that $\mathbf{c}$ bisects the angle between $\mathbf{a}$ and $\mathbf{b}$.

45. Prove Property 4 of the dot product. Use either the definition of a dot product (considering the cases $c > 0$, $c = 0$, and $c < 0$ separately) or the component form.

46. Suppose that all sides of a quadrilateral are equal in length and opposite sides are parallel. Use vector methods to show that the diagonals are perpendicular.

47. Prove the Cauchy-Schwarz Inequality:

$$|\mathbf{a} \cdot \mathbf{b}| \leq |\mathbf{a}| |\mathbf{b}|$$

48. The Triangle Inequality for vectors is

$$|\mathbf{a} + \mathbf{b}| \leq |\mathbf{a}| + |\mathbf{b}|$$

Use the Cauchy-Schwarz Inequality from Exercise 47 to prove the Triangle Inequality. [Hint: Use the fact that $|\mathbf{a} + \mathbf{b}|^2 = (\mathbf{a} + \mathbf{b}) \cdot (\mathbf{a} + \mathbf{b})$ and use Property 3 of the dot product.]

49. The Parallelogram Law states that

$$|\mathbf{a} + \mathbf{b}|^2 + |\mathbf{a} - \mathbf{b}|^2 = 2|\mathbf{a}|^2 + 2|\mathbf{b}|^2$$

- Give a geometric interpretation of the Parallelogram Law.
- Prove the Parallelogram Law. (See the hint in Exercise 48.)

50. Show that if $\mathbf{u} + \mathbf{v}$ and $\mathbf{u} - \mathbf{v}$ are orthogonal, then the vectors $\mathbf{u}$ and $\mathbf{v}$ must have the same length.

PROJECT Microarray Analysis of Genome Expression

BB

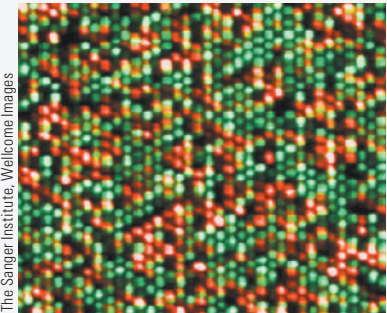


FIGURE 1
A microarray

Microarrays provide a technique for quantifying how the expression of different genes in the genome changes in response to changes in conditions. Suppose we are interested in how a drug affects genome expression. Cells are maintained under a reference, or normal, condition as well as in the presence of the drug. All gene products generated under the reference condition are labeled with a green dye, and all gene products generated in the presence of the drug are labeled with a red dye. The next step is to take all known genes individually and adhere copies of each to different locations on a glass plate. The dyed extracts from cells are then washed over the plate, and each of the dyed gene products binds with its corresponding gene.

The result is a glass plate containing thousands of colored dots, one for each gene. The dots vary in color from green to yellow to red. A dot is green if the corresponding gene tends to be expressed less in the presence of the drug than in the reference condition because most of the extract binding to this gene on the plate will have come from reference cells. A dot is yellow if the corresponding gene is expressed equally in both cell types, and it is red if the gene tends to be expressed more in the presence of the drug than in the reference condition. The entire collection of colored dots on the plate is called a microarray. An example is shown in Figure 1.

To analyze microarray data the color intensity for each gene is translated into a numerical score in the interval $(-\infty, \infty)$, with negative values indicating green and positive values indicating red. For simplicity, let's suppose there are five genes on the microarray and we have four different potential drugs, giving the following data:

	Gene 1	Gene 2	Gene 3	Gene 4	Gene 5
Drug A	1	3	5	6	9
Drug B	2	6	9	12	19
Drug C	-1	6	7	0	3
Drug D	5	2	5	7	12

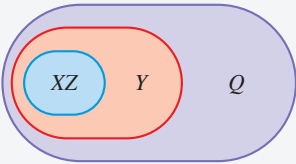


FIGURE 2
An example of hierarchical clustering. Drugs X and Z are most similar, followed by drug Y, and then drug Q.

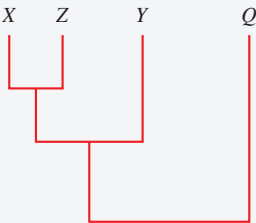


FIGURE 3
An example of a dendrogram corresponding to the clusters in Figure 2

Suppose our goal is to determine the similarity of drugs A, B, C, and D in terms of the patterns of genome expression that they induce. To do so, we need a measure of similarity of any two profiles.

1. Treating the expression profile for each drug as a point in $\mathbb{R}^5$, calculate the distances between all pairs of profiles.
2. The next step is to group the drugs hierarchically according to their similarity. To do so, first take the two drugs that are closest and draw a circle around them. Then, take the drug that is next closest to either of these first two and draw a curve that encloses all three (see Figure 2). Proceed until all drugs are included, giving a diagram depicting hierarchical clusters of decreasing similarity as shown in Figure 2. It is also possible to depict the drug similarities using a *dendrogram*, where the length of the path in the diagram between two drugs reflects the similarity between the drugs (see Figure 3). Construct both.
3. The other commonly used measure of similarity between profiles involves the dot product. Treating the expression profile for each drug as a vector in V_5 , calculate the cosine of the angle between each of the pairs of profiles as a measure of similarity. Repeat the clustering analysis in Problem 2 for this measure.

Problems 1 and 3 provide different measures of the similarity between pairs of profiles, and both are commonly used in microarray analysis. The similarity measure in Problem 1 is sometimes referred to as the Euclidean distance and the similarity measure in Problem 3 is sometimes referred to as the Pearson correlation coefficient.

PROJECT Vaccine Escape

BB



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Vaccine escape refers to the loss of effectiveness of a vaccine due to the antigenic evolution of the pathogen. Suppose that a vaccine protects against all pathogens within a circle of radius 5 centered at the origin in two-dimensional antigenic space. Further, suppose that the vector $\mathbf{v}$ giving the instantaneous velocity of evolution in antigenic space is always of fixed magnitude k and angle θ from a line connecting the pathogen to the origin, with $0 < \theta < \pi/2$. (See Figure 1.)

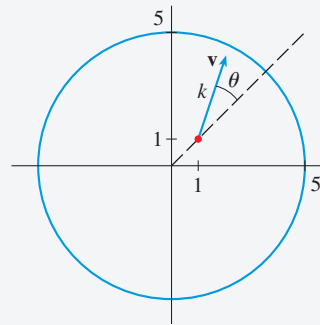


FIGURE 1

1. Suppose the pathogen is currently at location $(1, 1)$ in antigenic space. Using a vector projection, determine the vector that specifies the velocity at which the pathogen is evolving directly away from the origin.
2. Suppose the pathogen is currently at location $(1, 1)$ in antigenic space. Using a scalar projection, determine the speed at which the pathogen is evolving directly away from the origin.
3. In general, suppose the pathogen is at location (x, y) . Using a scalar projection, determine the speed at which the pathogen is evolving directly away from the origin.
4. Use your result from Problem 3 to write a differential equation that governs the distance of the pathogen from the origin.
5. Integrate the differential equation from Problem 4 and determine the time at which vaccine escape occurs (that is, the time when the pathogen escapes the vaccine's circle of protection), assuming the distance to the origin at time $t = 0$ is $\sqrt{2}$.

8.4 Matrix Algebra

Many of the examples from Sections 8.2 and 8.3 involve vectors that change over time; for example, antigenic evolution of the influenza virus or the heart voltage vector. In Section 8.5 we will consider how to model such changes using what is called a matrix model. Before doing so, however, we first introduce some ideas from matrix algebra. We will return to biological applications in Section 8.5.

Matrix Notation

A **matrix** is a rectangular array of numbers. We use uppercase symbols to denote matrices. The **size** of a matrix is defined by the number of rows and columns it contains. For

example, an $m \times n$ matrix has m rows and n columns. If A is an $m \times n$ matrix, then the ij th entry of A is the entry in the i th row and the j th column. We denote this entry by a_{ij} . For example, the matrix

$$(1) \quad A = \begin{bmatrix} 0 & 7 & 1 \\ 2 & 9 & 2 \end{bmatrix}$$

is a 2×3 matrix, and $a_{12} = 7$, $a_{21} = 2$, $a_{23} = 2$, and so forth.

A **square** matrix is one in which the number of rows is the same as the number of columns. We say that a square matrix is $n \times n$, or has size n , if it has n rows and columns.

The **transpose** of a matrix is obtained by interchanging its rows and columns and is denoted by a superscript T . For example, the transpose of the matrix in Equation 1, denoted by A^T , is

$$(2) \quad A^T = \begin{bmatrix} 0 & 2 \\ 7 & 9 \\ 1 & 2 \end{bmatrix}$$

If A is an $m \times n$ matrix, then A^T is an $n \times m$ matrix.

Vectors are often treated using the notation of matrices by introducing a distinction between those that are written as columns and those that are written as rows. All vectors we have seen so far have been written as **row vectors** in that their components are listed as a row. For example, we have been writing the vector with components x and y as $[x, y]$. Another possibility is to write this vector by placing its components in a column, giving the **column vector** $\begin{bmatrix} x \\ y \end{bmatrix}$. The row and column forms of a vector quantify the same thing but they are each used in different contexts in matrix algebra. We can view the row form of a vector as a $1 \times n$ matrix and the column form as an $n \times 1$ matrix. Furthermore, if $\mathbf{v}$ denotes a vector written in row form, then $\mathbf{v}^T$ is the same vector written in column form.

■ Matrix Addition and Scalar Multiplication

Only matrices of the same size can be added. If A and B are both $m \times n$ matrices with entries a_{ij} and b_{ij} , then $A + B$ is a new $m \times n$ matrix whose entries are $a_{ij} + b_{ij}$. Thus the matrix sum $A + B$ is calculated by adding the corresponding entries of each matrix.

EXAMPLE 1 | Evaluate the following sums, if possible.

$$(a) \quad M + N, \quad \text{where } M = \begin{bmatrix} 2 & x & 9 \\ 4 & 5 & 6 \end{bmatrix} \text{ and } N = \begin{bmatrix} 92 & 6 & 2 \\ 15 & 3 & 1 \end{bmatrix}$$

$$(b) \quad X + Y, \quad \text{where } X = \begin{bmatrix} 5 & 3 \\ 7 & 13 \end{bmatrix} \text{ and } Y = \begin{bmatrix} 3 & 9 & 21 \\ 5 & 7 & 6 \end{bmatrix}$$

SOLUTION

(a) Both matrices are 2×3 and therefore can be added. Adding the corresponding entries gives $M + N = \begin{bmatrix} 94 & x + 6 & 11 \\ 19 & 8 & 7 \end{bmatrix}$.

(b) Matrix X is 2×2 while Y is 2×3 . Since they are not the same size, they cannot be added. ■

Scalar multiplication with matrices works just as it does with vectors. If A is an $m \times n$ matrix with entries a_{ij} and c is a scalar, then the product cA is an $m \times n$ matrix with entries ca_{ij} . In other words, the product cA is calculated by multiplying each entry of A by c .

Matrix subtraction can be defined through a combination of scalar multiplication and matrix addition. If A and B are both $m \times n$ matrices with entries a_{ij} and b_{ij} , then $A - B$ is calculated by first multiplying B by -1 and then adding this to A . Thus the difference $A - B$ is calculated by subtracting entry b_{ij} from a_{ij} . Again notice that matrix subtraction can be performed only with matrices of the same size.

Finally, two matrices A and B are said to be **equal** if $A - B = 0$ where 0 is an $m \times n$ matrix of zeros.

Properties of Matrix Addition If A , B , and C are $m \times n$ matrices and a and b are scalars, then

- | | |
|-------------------------|--------------------------------|
| 1. $A + B = B + A$ | 2. $A + (B + C) = (A + B) + C$ |
| 3. $A + 0 = A$ | 4. $A + (-A) = 0$ |
| 5. $a(A + B) = aA + aB$ | 6. $(a + b)A = aA + bA$ |

Properties 1–6 can be verified using the definitions of matrix addition and multiplication by a scalar.

Matrix Multiplication

Some matrices can also be multiplied with one another. Matrix multiplication can be viewed as an extension of the dot product of vectors. If we wish to calculate the matrix product AB , we view the matrix A as a collection of row vectors and B as a collection of column vectors. The ij th entry of the resulting product is then the dot product of the i th row of A with the j th column of B . (See Figure 1.) For example, if A is a 2×3 matrix and B is a 3×2 matrix, we have

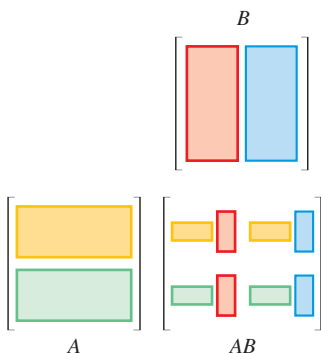


FIGURE 1

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} & a_{11}b_{12} + a_{12}b_{22} + a_{13}b_{32} \\ a_{21}b_{11} + a_{22}b_{21} + a_{23}b_{31} & a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32} \end{bmatrix}$$

Note that the resulting matrix is 2×2 . More generally, if A is an $m \times n$ matrix and B is an $n \times p$ matrix, then $C = AB$ is an $m \times p$ matrix whose ij th entry is found by taking the dot product of the i th row of A with the j th column of B , as shown in Figure 2.

FIGURE 2

The i th row of A is indicated by $\mathbf{r}_i$.
The j th column of B is indicated by $\mathbf{c}_j$.

$$\begin{bmatrix} \text{---} & \mathbf{r}_1 & \text{---} \\ \text{---} & \mathbf{r}_2 & \text{---} \\ \vdots & \vdots & \vdots \\ \text{---} & \mathbf{r}_m & \text{---} \end{bmatrix} \begin{bmatrix} | & | & \cdots & | \\ \mathbf{c}_1 & \mathbf{c}_2 & \cdots & \mathbf{c}_p \\ | & | & \cdots & | \end{bmatrix} = \begin{bmatrix} \mathbf{r}_1 \cdot \mathbf{c}_1 & \mathbf{r}_1 \cdot \mathbf{c}_2 & \cdots & \mathbf{r}_1 \cdot \mathbf{c}_p \\ \mathbf{r}_2 \cdot \mathbf{c}_1 & \mathbf{r}_2 \cdot \mathbf{c}_2 & \cdots & \mathbf{r}_2 \cdot \mathbf{c}_p \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{r}_m \cdot \mathbf{c}_1 & \mathbf{r}_m \cdot \mathbf{c}_2 & \cdots & \mathbf{r}_m \cdot \mathbf{c}_p \end{bmatrix}$$

$A \qquad B \qquad AB$

Matrix multiplication is not defined for matrices where the number of columns of the first matrix is different from the number of rows of the second matrix. A simple way to determine if a given matrix multiplication is defined is to write the size of the first matrix,

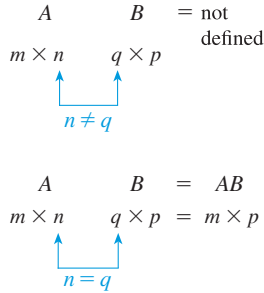


FIGURE 3

followed by the size of the second matrix (see Figure 3). If the two “inner” numbers are not the same, then the matrix multiplication is not defined. If they are the same, then the resulting matrix has size given by the two “outer” numbers as illustrated in Figure 3.

Matrix multiplication is summarized by the following rule.

Matrix Multiplication If A is an $m \times n$ matrix and B is an $n \times p$ matrix, then their product $C = AB$ is an $m \times p$ matrix whose entries are given by

$$c_{ij} = \sum_{k=1}^n a_{ik}b_{kj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$

for $1 \leq i \leq m$ and $1 \leq j \leq p$.

EXAMPLE 2 | Determine each matrix product if it is defined.

$$A = \begin{bmatrix} 2 & 7 \\ 9 & -3 \end{bmatrix} \quad B = \begin{bmatrix} 3 & -7 & 2 \\ 1 & 5 & 9 \end{bmatrix} \quad C = \begin{bmatrix} 5 & -6 \\ 8 & 2 \end{bmatrix}$$

- (a) AB (b) BA (c) AC (d) CA

SOLUTION

(a) Matrices A and B have sizes 2×2 and 2×3 , respectively. From Figure 3, since $n = q = 2$, matrix multiplication is therefore defined and the resulting matrix is 2×3 . Performing the calculation, we obtain

$$\begin{bmatrix} 2 & 7 \\ 9 & -3 \end{bmatrix} \begin{bmatrix} 3 & -7 & 2 \\ 1 & 5 & 9 \end{bmatrix} = \begin{bmatrix} 6 + 7 & -14 + 35 & 4 + 63 \\ 27 - 3 & -63 - 15 & 18 - 27 \end{bmatrix} = \begin{bmatrix} 13 & 21 & 67 \\ 24 & -78 & -9 \end{bmatrix}$$

(b) Matrices B and A have sizes 2×3 and 2×2 , respectively. From Figure 3, since $n = 3$ and $q = 2$, $n \neq q$ and the product is not defined.

(c) Using the rule from Figure 3, we have $n = q = 2$. The resulting matrix is 2×2 . Performing the calculation, we obtain

$$\begin{bmatrix} 2 & 7 \\ 9 & -3 \end{bmatrix} \begin{bmatrix} 5 & -6 \\ 8 & 2 \end{bmatrix} = \begin{bmatrix} 10 + 56 & -12 + 14 \\ 45 - 24 & -54 - 6 \end{bmatrix} = \begin{bmatrix} 66 & 2 \\ 21 & -60 \end{bmatrix}$$

(d) Using the rule from Figure 3, we have $n = q = 2$. The resulting matrix is 2×2 and, performing the calculation, we obtain

$$\begin{bmatrix} 5 & -6 \\ 8 & 2 \end{bmatrix} \begin{bmatrix} 2 & 7 \\ 9 & -3 \end{bmatrix} = \begin{bmatrix} 10 - 54 & 35 + 18 \\ 16 + 18 & 56 - 6 \end{bmatrix} = \begin{bmatrix} -44 & 53 \\ 34 & 50 \end{bmatrix}$$

Multiplication of two quantities α and β is said to be **commutative** if $\alpha\beta = \beta\alpha$. Parts (c) and (d) of Example 2 illustrate the important fact that matrix multiplication is not, in general, commutative ($AC \neq CA$ in this example).

An $n \times n$ matrix D is called **diagonal** if all off-diagonal entries are zero; that is, $d_{ij} = 0$ for all $i \neq j$. Multiplication of a diagonal matrix with itself is especially easy.

EXAMPLE 3 | For an arbitrary 2×2 diagonal matrix D with entries d_{ii} , calculate the matrix DD .

SOLUTION Calculating the matrix product, we obtain

$$DD = \begin{bmatrix} d_{11} & 0 \\ 0 & d_{22} \end{bmatrix} \begin{bmatrix} d_{11} & 0 \\ 0 & d_{22} \end{bmatrix} = \begin{bmatrix} d_{11}^2 & 0 \\ 0 & d_{22}^2 \end{bmatrix}$$

Thus DD is a diagonal matrix with entries d_{ii}^2 . ■

An alternative notation for the matrix product MM is M^2 . Likewise, M^k represents the matrix M multiplied by itself k times. It is referred to as the k th power of M . Exercise 7 shows that the simple pattern for the second power of a diagonal matrix illustrated in Example 3 also holds for larger matrices as well as for higher matrix powers.

A diagonal matrix is called an **identity** matrix if all the entries on the diagonal are 1. Identity matrices are usually denoted by I and play the same role in matrix multiplication that the number 1 plays in regular multiplication.

EXAMPLE 4 | For an arbitrary 2×2 matrix A show that $AI = IA = A$.

SOLUTION Performing the required matrix products we obtain

$$AI = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = A$$

and
$$IA = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = A$$
 ■

We finish by summarizing some important properties of matrix multiplication. These can be verified using the definitions presented in this section (see Exercise 11).

Properties of Matrix Multiplication Suppose A , B , and C are matrices and a and b are scalars. Provided the required matrix multiplications are defined, then

- | | |
|-------------------------|-------------------------|
| 1. $A(BC) = (AB)C$ | 2. $(aA)(bB) = abAB$ |
| 3. $A(B + C) = AB + AC$ | 4. $(B + C)A = BA + CA$ |
| 5. $IA = A, AI = A$ | 6. $0A = 0, A0 = 0$ |

Note: Matrix multiplication is *not*, in general, commutative; that is, $AB \neq BA$.

EXERCISES 8.4

1–2 Suppose A , B , and C are 2×2 matrices, E , F , and G are 3×3 matrices, H and K are 2×3 matrices, and L and M are 3×2 matrices. For each of the following, if the operation is defined, specify the size of the matrix that results.

- | | |
|---------------------|------------------------|
| 1. (a) $A + 4C$ | (b) $\frac{1}{2}K + L$ |
| (c) $5K + 3H$ | (d) $0G + 3(E + F)$ |
| (e) $3A + 6(B + M)$ | (f) $12M - L$ |

- (g) $F + G - 2C$
 (h) $\alpha F + \beta G$, where α and β are scalars

- | | |
|-----------------|-----------------|
| 2. (a) $AB + C$ | (b) $3GF$ |
| (c) $CK + B$ | (d) $CK + H$ |
| (e) EMC | (f) GLH |
| (g) HLG | (h) $2EL + 5MB$ |

3. Using the given matrices, calculate the quantities, if they are defined.

$$A = \begin{bmatrix} 2 & 5 \\ 1 & 7 \end{bmatrix} \quad B = \begin{bmatrix} 7 & x \\ a & 5 \end{bmatrix} \quad C = \begin{bmatrix} 9 & 2 \\ 7 & 10 \end{bmatrix}$$

$$E = \begin{bmatrix} 0 & 3 & 1 \\ 7 & 6 & 0 \\ 9 & 13 & 5 \end{bmatrix} \quad F = \begin{bmatrix} x & 2 & 9 \\ 6 & y & 13 \\ 0 & 1 & 0 \end{bmatrix}$$

$$G = \begin{bmatrix} 13 & 2 & 0 \\ 5 & 9 & 12 \\ 7 & 0 & 1 \end{bmatrix}$$

$$H = \begin{bmatrix} 3 & 1 & 7 \\ 15 & 0 & 2 \end{bmatrix} \quad K = \begin{bmatrix} 1 & 0 & 1 \\ 0 & y & 1 \end{bmatrix}$$

- (a) $A - 3C$ (b) $3F + G - E$
 (c) $5K + 9H$ (d) $H - 12G$
 (e) $5B - A$ (f) $B - 9K$
 (g) $3F - F$ (h) $H - K + 2H$
4. Using the given matrices, calculate the quantities, if they are defined.

$$A = \begin{bmatrix} 7 & 0 \\ 1 & 9 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 2 \\ 13 & 6 \end{bmatrix} \quad C = \begin{bmatrix} 15 & 0 \\ 0 & 2 \end{bmatrix}$$

$$E = \begin{bmatrix} x & 5 & 3 \\ 0 & 9 & 1 \\ 6 & 4 & 0 \end{bmatrix} \quad F = \begin{bmatrix} y & 0 & y \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$$G = \begin{bmatrix} 1 & 2 & 3 \\ 5 & 4 & 6 \\ 9 & 7 & 8 \end{bmatrix}$$

$$H = \begin{bmatrix} 2 & 1 & 3 \\ 3 & 7 & 6 \end{bmatrix} \quad K = \begin{bmatrix} 9 & 2 \\ 1 & 5 \\ 7 & 8 \end{bmatrix}$$

- (a) $A - 2BC$ (b) EG
 (c) $FK - 2K$ (d) HEA
 (e) $3BC - HK$ (f) $7K - H$
 (g) $AHKB$ (h) $BKHA$
5. Find the transpose of each matrix.

(a) $\begin{bmatrix} 3X \\ 1 \\ 2 \end{bmatrix}$

(b) $\begin{bmatrix} 3 & 3 & 9 \end{bmatrix}$

(c) $\begin{bmatrix} 2 & 1 & 7 \\ 8 & 3 & 6 \end{bmatrix}$

(d) $\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$

6–10 The notation A^k means the matrix A multiplied with itself k times.

6. (a) For the 2×2 identity matrix I , show that $I^2 = I$.
 (b) For the $n \times n$ identity matrix I , show that $I^2 = I$.
 (c) What do you think the entries of I^k are?

7. (a) Show that, for all $n \times n$ diagonal matrices D having entries d_{ii} , the matrix D^2 is also diagonal with entries d_{ii}^2 .
 (b) For an arbitrary $n \times n$ diagonal matrix D , what do you think the entries of D^k are?

8. Show that, in general, if A is a 2×2 matrix, then

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}^2 \neq \begin{bmatrix} a_{11}^2 & a_{12}^2 \\ a_{21}^2 & a_{22}^2 \end{bmatrix}$$

9. Consider the matrix $C = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.

- (a) Calculate C^2 , C^3 , C^4 , and C^5 .
 (b) What do you think C^k is?

10. Find an example of a nonzero 2×2 matrix whose square is the zero matrix.

11. For arbitrary 2×2 matrices A , B , and C , verify the following properties.

- (a) $A(BC) = (AB)C$
 (b) $A(B + C) = AB + AC$
 (c) $(B + C)A = BA + CA$

12. Suppose that A and B are arbitrary 2×2 matrices. Is the quantity given always equal to $(A + B)^2$?

- (a) $A^2 + 2AB + B^2$
 (b) $(B + A)^2$
 (c) $A(A + B) + B(A + B)$
 (d) $A^2 + AB + BA + B^2$

13. Suppose that A is a 2×2 matrix that commutes with all possible 2×2 matrices. Show that, necessarily, $a_{11} = a_{22}$ and $a_{12} = a_{21} = 0$. *Hint:* Try using

$$X = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad Y = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

as two particular 2×2 matrices with which A must commute.

14. (a) For any 2×2 matrices A and B show that

$$(A + B)^T = A^T + B^T$$

- (b) For any $n \times n$ matrices A and B show that

$$(A + B)^T = A^T + B^T$$

15. (a) For any 2×2 matrices A and B show that

$$(AB)^T = B^T A^T$$

- (b) For any $n \times n$ matrices A and B show that

$$(AB)^T = B^T A^T$$

16. A matrix A is called **symmetric** if $A^T = A$. Verify, for all 3×3 matrices B , that $B + B^T$ is symmetric.
17. Suppose a matrix A satisfies the equation $A + A^T = 0$. What must be true about the entries of A ? Such matrices are called **skew-symmetric**.
18. An **upper triangular matrix** is a square matrix with all zeros below its diagonal. A **lower triangular matrix** is a square matrix with all zeros above its diagonal.
- (a) For all 3×3 upper triangular matrices U , verify that U^2 is also upper triangular.
- (b) For all 3×3 lower triangular matrices L , verify that L^2 is also lower triangular.

8.5 Matrices and the Dynamics of Vectors

In this section we show how matrices can be used to model changes in a vector from one point in time to the next. Such models are called matrix models.

■ Systems of Difference Equations: Matrix Models

In Sections 1.6 and 2.1 we studied the following difference equation for population size:

$$(1) \quad n_{t+1} = Rn_t$$

Note: We have changed notation slightly from Sections 1.6 and 2.1—we are now using a lowercase n to denote population size.

Equation 1 has an equilibrium at $\hat{n} = 0$. For other values of n we saw that n grows in magnitude if $R > 1$ and decays to 0 if $0 < R < 1$. In the special case where $R = 1$, any value of n is an equilibrium because Equation 1 then becomes $n_{t+1} = n_t$.

Suppose we want to model the population size when there are adults and juveniles. Each adult produces two juveniles in one time step, and half the juveniles survive to become adults. Finally, $\frac{1}{3}$ of adults survive from one time step to the next. Using j_t and a_t to denote the number of juveniles and adults at time t , we obtain the *system of difference equations*

$$(2) \quad \begin{aligned} j_{t+1} &= 2a_t \\ a_{t+1} &= \frac{1}{2}j_t + \frac{1}{3}a_t \end{aligned}$$

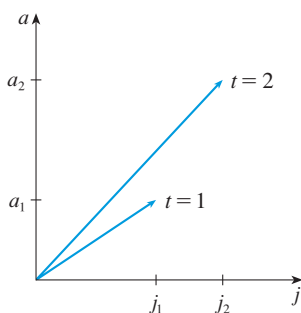


FIGURE 1

We can view the state of the population at any time t as a vector $\begin{bmatrix} j_t \\ a_t \end{bmatrix}$ (see Figure 1).

We have written this as a column vector in anticipation of using the techniques of matrix algebra from Section 8.4 to simplify the notation. In particular, let's view this vector as a 2×1 matrix and consider calculating the following matrix product using the rules of matrix multiplication from Section 8.4:

$$\begin{bmatrix} 0 & 2 \\ \frac{1}{2} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} j_t \\ a_t \end{bmatrix}$$

This is the product of a 2×2 matrix with a 2×1 matrix, and so matrix multiplication is defined. From the rule in Figure 8.4.3, we anticipate the result to be a 2×1 matrix (that is, a column vector).

Performing the multiplication, we obtain

$$\begin{bmatrix} 0 & 2 \\ \frac{1}{2} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} j_t \\ a_t \end{bmatrix} = \begin{bmatrix} 0 \cdot j_t + 2a_t \\ \frac{1}{2}j_t + \frac{1}{3}a_t \end{bmatrix} = \begin{bmatrix} 2a_t \\ \frac{1}{2}j_t + \frac{1}{3}a_t \end{bmatrix}$$

From Equations 2 we see that the resulting vector can be written as $\begin{bmatrix} j_{t+1} \\ a_{t+1} \end{bmatrix}$, and therefore the system of difference equations in (2) can be written in matrix notation as

$$\begin{bmatrix} j_{t+1} \\ a_{t+1} \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ \frac{1}{2} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} j_t \\ a_t \end{bmatrix}$$

As a final step we define Q to be the matrix $\begin{bmatrix} 0 & 2 \\ \frac{1}{2} & \frac{1}{3} \end{bmatrix}$ and $\mathbf{n}_t$ to be the column vector $\begin{bmatrix} j_t \\ a_t \end{bmatrix}$. We then have

$$(3) \quad \mathbf{n}_{t+1} = Q\mathbf{n}_t$$

The parallel between the single-variable recursion in Equation 1 for n_t and Equation 3 for the vector $\mathbf{n}_t$ is now apparent. The change in the vector $\begin{bmatrix} j_t \\ a_t \end{bmatrix}$ is described by the product of the matrix of constants Q with this vector, just as the change in n_t in Equation 1 is described by the product of the constant R with n_t . In the vector setting the growth factor R in Equation 1 is replaced by the matrix Q in Equation 3. Models such as Equation 3 are called **matrix models**.

As with Equation 1, how the vector $\mathbf{n}_t$ changes over time depends on the initial condition and on the matrix of constants Q . And as with Equation 1, we can recursively apply Equation 3 to an initial vector to generate numerical solutions. Figure 2 illustrates an example of this for Equations 2. Both j_t and a_t initially oscillate, but the population size of both juveniles and adults appears to grow in the long run. In later sections of this chapter we will learn how to determine the behavior of such matrix models mathematically. As with Equation 1, the long-term behavior of the model depends on a measure of the magnitude of the “growth factor” Q .

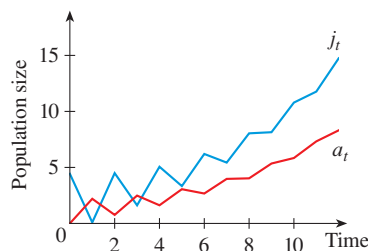


FIGURE 2



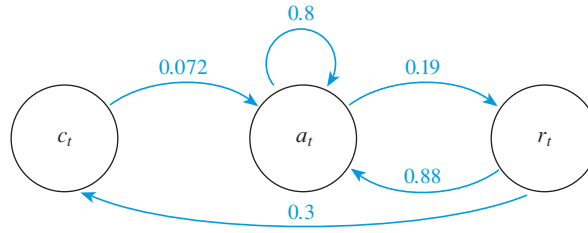
FIGURE 3
A right whale

EXAMPLE 1 | **BB** Conservation biology of right whales Matrix models are often used to predict the population size of endangered species such as right whales. Right whales have important stage structure to their life cycle. A simplified model considers three stages: (i) newborn calves, (ii) mature but nonreproducing adults, and (iii) reproductive adults. Let's use c_t , a_t , and r_t , respectively, to denote the number of individuals in each of these stages. Suppose that newborn calves mature into adults with probability 0.072 each time step (otherwise they die). Nonreproducing adults remain that way with probability 0.8 or begin reproducing with probability 0.19 (otherwise they die). Reproducing individuals produce, on average, 0.3 calves each time step and revert to the nonreproducing adult class with probability 0.88 (otherwise they die). Construct the matrix model for this population.

SOLUTION First we summarize the model using what is called a *matrix diagram* in Figure 4.

FIGURE 4

Each circle represents a life stage. An arrow is drawn from circle j to i if stage j contributes to the number of individuals in stage i in the next time step. The numbers next to the arrows indicate the *per capita* contribution.



We proceed by deriving a difference equation for each of the three variables in turn. From Figure 4 the number of calves at time $t + 1$ is the per capita number produced by each reproductive individual, 0.3, multiplied by the number of these individuals at time t . Therefore, $c_{t+1} = 0.3r_t$. The number of adults at time $t + 1$ is the number of calves at time t multiplied by the per capita probability that they mature, 0.072, plus the number of adults at time t multiplied by the per capita probability that they remain as nonreproducing adults, 0.8, plus the number of reproductive individuals at time t multiplied by the per capita probability that they revert to the nonreproducing adult stage, 0.88. This gives

$$a_{t+1} = 0.072c_t + 0.8a_t + 0.88r_t$$

Finally, the number of reproductive individuals at time $t + 1$ is the number of adults at time t multiplied by the per capita probability that they move into the reproductive class, 0.19, giving $r_{t+1} = 0.19a_t$. You can verify that organizing this using matrix notation gives

$$\begin{bmatrix} c_{t+1} \\ a_{t+1} \\ r_{t+1} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0.3 \\ 0.072 & 0.8 & 0.88 \\ 0 & 0.19 & 0 \end{bmatrix} \begin{bmatrix} c_t \\ a_t \\ r_t \end{bmatrix}$$

Example 1 introduces the idea of drawing a **matrix diagram** to organize the construction of a matrix model. In general, such diagrams consists of a single circle for each variable of the model, and an arrow from circle j to i whenever type j contributes to the amount of type i in the next time step (see Figure 5). The corresponding matrix can then be written directly from the diagram by noting that the entry in the i th row and j th column of the matrix corresponds to the arrow from circle j to i .

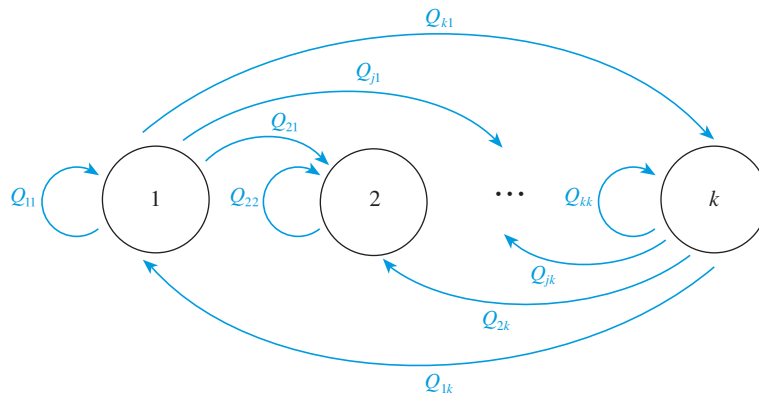


FIGURE 5

A matrix diagram

EXAMPLE 2 | Breast cancer Consider a population of patients at risk of developing breast cancer. Each individual's health status can be in one of three states: (1) healthy, (2) early-stage disease, or (3) late-stage disease. Each year 0.2% of healthy females develop early-stage disease, with the rest remaining healthy. Likewise, 45% of women with early-stage disease recover through treatment, 45% develop late-stage disease, and 10% remain in the early-stage category. All individuals with late-stage disease remain in that state.

- (a) Using h_t , e_t , and l_t to denote the number of individuals in each category, construct the matrix diagram for this example.
 (b) Construct the matrix model corresponding to the diagram from part (a).

SOLUTION

- (a) Each of the three states is denoted by a circle in Figure 6, and we draw arrows corresponding to the possible transitions in health status. Finally, we label each arrow with its appropriate value.

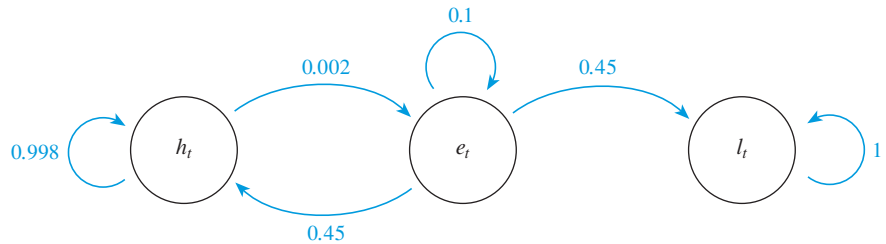


FIGURE 6

- (b) We find the entry in the i th row and j th column of the matrix by looking for an arrow from circle j to circle i in Figure 6, giving

$$\begin{bmatrix} h_{t+1} \\ e_{t+1} \\ l_{t+1} \end{bmatrix} = \begin{bmatrix} 0.998 & 0.45 & 0 \\ 0.002 & 0.1 & 0 \\ 0 & 0.45 & 1 \end{bmatrix} \begin{bmatrix} h_t \\ e_t \\ l_t \end{bmatrix}$$

■ Leslie Matrices

Matrix models are often used to model the size of age-structured populations. Consider a simplified model for a Coho salmon population. Individuals hatch from eggs in rivers on the West Coast of North America and migrate to the ocean. At around three years of age they migrate back to the river, reproduce, and die. Let's use $n_{1,t}$, $n_{2,t}$, and $n_{3,t}$ to denote the number of individuals in each age class at time t , where time is measured in years. Figure 7 gives the matrix diagram for the model, along with estimated values for the constants.

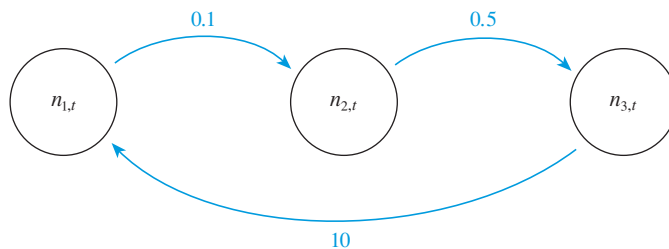


FIGURE 7

The resulting matrix model is

$$\begin{bmatrix} n_{1,t+1} \\ n_{2,t+1} \\ n_{3,t+1} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 10 \\ 0.1 & 0 & 0 \\ 0 & 0.5 & 0 \end{bmatrix} \begin{bmatrix} n_{1,t} \\ n_{2,t} \\ n_{3,t} \end{bmatrix}$$

Figure 8 shows the number of individuals in each age class over time. The graph looks rather complicated, but the population size appears to decay to zero in the long run.

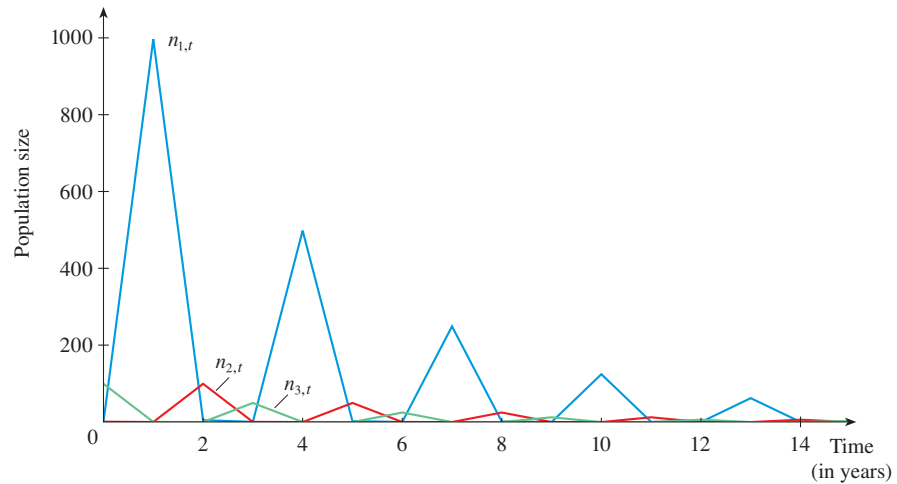


FIGURE 8

Leslie

Patrick H. Leslie was a mathematical biologist who worked at Oxford University and pioneered the use of matrix models in population biology in the 1940s.

Matrices for age-structured models like that of the Coho salmon always have a special structure. They are called **Leslie matrices** after Patrick H. Leslie and are often denoted by the symbol L . To see this structure better, let's examine the general case of k age classes. Using b_i for the number of offspring produced by an age i individual and s_i for the probability that an age i individual survives to become an age $i + 1$ individual, we have the matrix diagram shown in Figure 9.

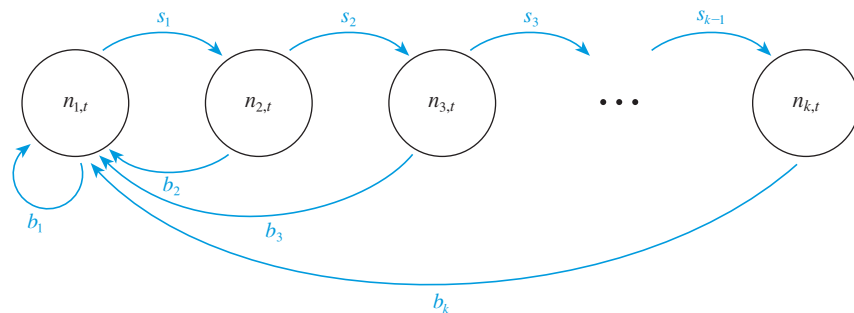


FIGURE 9

Algebraically, we have

$$n_{1,t+1} = b_1 n_{1,t} + b_2 n_{2,t} + \cdots + b_k n_{k,t}$$

$$n_{2,t+1} = s_1 n_{1,t}$$

$$n_{3,t+1} = s_2 n_{2,t}$$

$$\vdots$$

$$n_{k,t+1} = s_{k-1} n_{k-1,t}$$

or, in matrix notation,

$$\begin{bmatrix} n_{1,t+1} \\ n_{2,t+1} \\ n_{3,t+1} \\ \vdots \\ n_{k,t+1} \end{bmatrix} = \begin{bmatrix} b_1 & b_2 & b_3 & \cdots & b_k \\ s_1 & 0 & 0 & \cdots & 0 \\ 0 & s_2 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & s_{k-1} & 0 \end{bmatrix} \begin{bmatrix} n_{1,t} \\ n_{2,t} \\ n_{3,t} \\ \vdots \\ n_{k,t} \end{bmatrix}$$

Again we have assumed that a single time step corresponds to the time it takes to move from one age class to the next. As a result, all surviving individuals must progress from one age class to the next each time step, giving the matrix a special form. The age-specific birth rates make up the first row of the matrix, and the age-specific survival probabilities lie along the *subdiagonal* of the matrix (that is, the diagonal lying immediately below the diagonal that runs from the top left to the bottom right corner of the matrix).

Summary

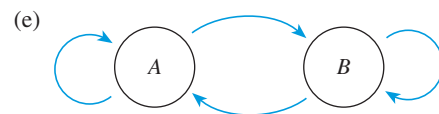
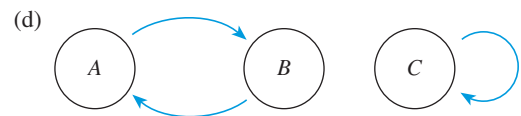
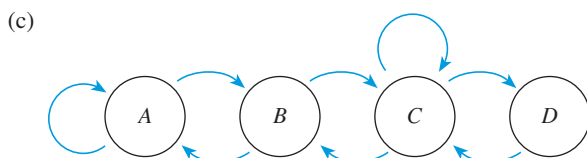
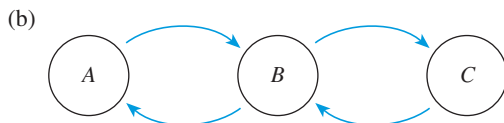
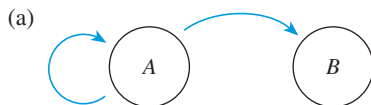
Matrix models are a special kind of dynamical system that govern how vectors change from one time step to the next. They consist of a column vector of variables, $\mathbf{n}_t$, that changes from one time step to the next, along with a matrix of constants, A , that determines how the vector changes. They take the form

$$\mathbf{n}_{t+1} = A\mathbf{n}_t$$

For any given initial vector we can recursively apply the equation to obtain numerical solutions. As we will see in Section 8.8, we can also analyze such equations quite extensively from a mathematical perspective and thereby gain a complete understanding of how the vector changes through time.

EXERCISES 8.5

- Construct the corresponding matrix for each of the following matrix diagrams by placing an X as the ij th entry of the matrix if the entry is nonzero, and a 0 otherwise.



- For each of the following matrices an X in entry ij represents the fact that type j contributes to type i in the next time step. Construct the corresponding matrix diagram for each.

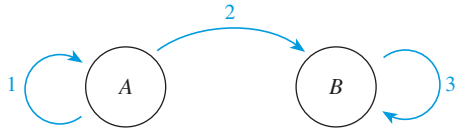
(a) $\begin{bmatrix} X & X \\ 0 & X \end{bmatrix}$ (b) $\begin{bmatrix} X & 0 & X \\ 0 & 0 & X \\ X & 0 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 0 & X & 0 & X \\ X & 0 & X & 0 \\ X & X & 0 & 0 \\ 0 & 0 & X & X \end{bmatrix}$ (d) $\begin{bmatrix} X & 0 & 0 \\ 0 & X & 0 \\ 0 & 0 & X \end{bmatrix}$

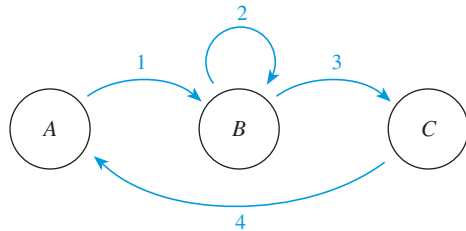
(e) $\begin{bmatrix} 0 & X \\ X & 0 \end{bmatrix}$

3–4 Construct the corresponding matrix for each of the following matrix diagrams.

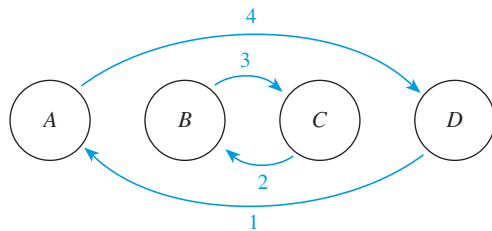
3. (a)



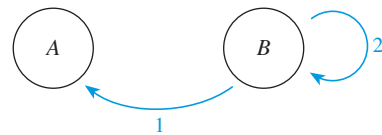
(b)



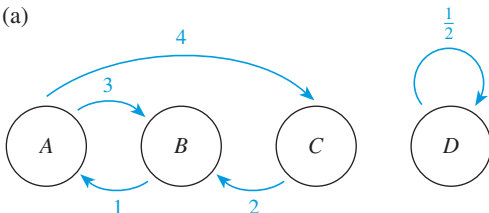
(c)



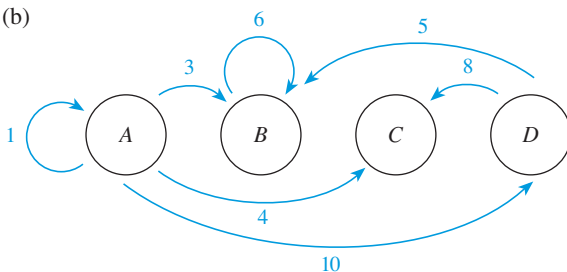
(d)



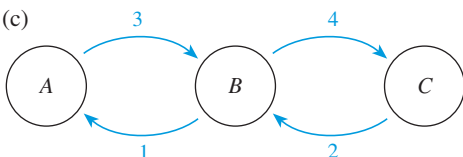
4. (a)



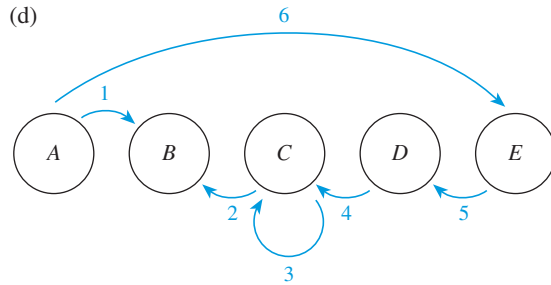
(b)



(c)



(d)



5–6 Each of the following matrices represents a system of difference equations. Construct the corresponding matrix diagram.

5. (a) $\begin{bmatrix} 1 & 6 \\ 5 & 0 \end{bmatrix}$

(b) $\begin{bmatrix} 2 & 0 & 3 \\ 5 & 1 & 2 \\ 7 & 9 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

(d) $\begin{bmatrix} 7 & 6 & 0 \\ 0 & 2 & 3 \\ 0 & 1 & 0 \end{bmatrix}$

6. (a) $\begin{bmatrix} 3 & 0 & 7 & 9 & 11 \\ 0.1 & 0.1 & 0.8 & 0 & 0 \\ 9 & 0 & 1 & 1 & 1 \\ 6 & 5 & 4 & 3 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

(b) $\begin{bmatrix} 0 & 1 & 3 & 0 \\ 3 & 1 & 3 & 1 \\ 0 & 0 & 9 & 0 \\ 7 & 0 & 0 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 0 & 0 \\ 0.25 & 0.5 & 0.25 \\ 0 & 0 & 1 \end{bmatrix}$

(d) $\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$

7. Suppose the matrix representing a system of difference equations contains a row of zeros. What does this imply about the system?

8. Suppose the matrix representing a system of difference equations contains a column of zeros. What does this imply about the system?

9–15 Construct the matrix diagram for each of the following models.

9. Dispersal A population of dragonflies occupies two ponds. Each day 20% of the individuals in pond A move to pond B, and 30% of the individuals in pond B move to pond

A. Using A_t and B_t to denote the number of individuals in each pond at time t , we have

$$A_{t+1} = 0.8A_t + 0.3B_t$$

$$B_{t+1} = 0.2A_t + 0.7B_t$$

- 10. Mutation** Each generation 5% of individuals carrying allele X mutate to carry allele Y. Using X_t and Y_t to denote the number of individuals carrying each allele at time t , we have

$$X_{t+1} = 0.95X_t$$

$$Y_{t+1} = 0.05X_t + Y_t$$

- 11. Cancer progression** Consider a population of women at risk of developing breast cancer. Each individual's health status can be in one of three states: (1) healthy, (2) early-stage disease, or (3) late-stage disease. Each year 0.2% of healthy females develop early-stage disease, with the rest remaining healthy. Likewise, 65% of women with early-stage disease recover through treatment and 35% develop late-stage disease. Finally, 10% of individuals with late-stage disease revert to early stage disease while the rest remain in late stage. Using h_t , e_t , and l_t to denote the number of individuals in each state at time t , we have

$$h_{t+1} = 0.998h_t + 0.65e_t$$

$$e_{t+1} = 0.002h_t + 0.1l_t$$

$$l_{t+1} = 0.35e_t + 0.9l_t$$

- 12. Yellow perch** Yellow perch is a species of freshwater fish. Suppose a population of interest lives to a maximum age of 4 years. Only 5% of age 1 individuals survive to age 2, 20% of age 2 individuals survive to age 3, and 75% of age 3 individuals survive to age 4. Both age 3 and age 4 individuals reproduce, producing 100 and 150 age 1 offspring, respectively. Using $n_{i,t}$ for the number of individuals of age i at time t , we have

$$n_{1,t+1} = 100n_{3,t} + 150n_{4,t}$$

$$n_{2,t+1} = 0.05n_{1,t}$$

$$n_{3,t+1} = 0.2n_{2,t}$$

$$n_{4,t+1} = 0.75n_{3,t}$$

- 13. Drug diffusion** A drug taken orally eventually diffuses into the bloodstream through the stomach. Let s_t be the amount of drug in the stomach at time t and b_t the amount that has been absorbed into the bloodstream. Each time step, 50% of the drug is absorbed from the stomach into the bloodstream and 80% of the drug in the bloodstream is metabolized. We have

$$s_{t+1} = 0.5s_t$$

$$b_{t+1} = 0.5s_t + 0.2b_t$$

- 14. Breeding systems** Cooperatively breeding bird species are those in which some individuals forgo their own

reproduction to help others reproduce. Therefore the population consists of three kinds of individuals: juveniles, helpers, and reproductives. Using j_t , h_t , and r_t to denote the number of each kind at time t , a model for the population size of each is

$$j_{t+1} = 2r_t$$

$$h_{t+1} = 0.9j_t + 0.8h_t + 0.1r_t$$

$$r_{t+1} = 0.1j_t + 0.2h_t + 0.9r_t$$

- 15. Ecological succession** is a process by which a biological community progresses between different states. Suppose the community can be either primarily grassland, primarily shrub, or primarily forest, and this can change from year to year. Using g_t , s_t , and f_t to denote the probability that the community is in each of the three possible states at time t , we have

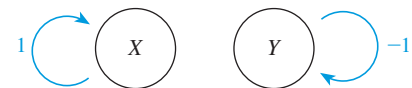
$$g_{t+1} = 0.9g_t + 0.001s_t$$

$$s_{t+1} = 0.1g_t + 0.799s_t + 0.001f_t$$

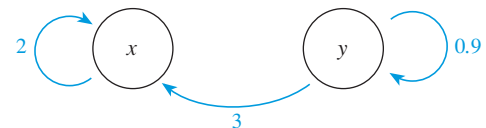
$$f_{t+1} = 0.2s_t + 0.999f_t$$

- 16–19** Write the system of difference equations and construct the corresponding matrix that describes this system.

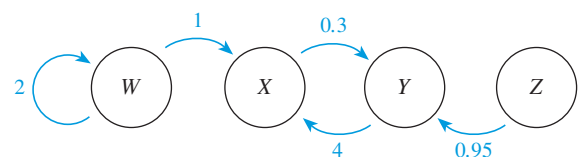
- 16. Vectorcardiography** Suppose the components X and Y of the heart's voltage vector change from one beat to the next according to the diagram



- 17. Antigenic evolution** Suppose influenza viruses are plotted in two-dimensional antigenic space, and that the x - and y -coordinates change from one year to the next according to the diagram

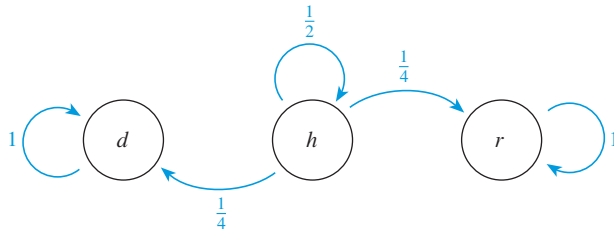


- 18. Genome expression profiles** A genome expression profile can be characterized as a vector. Consider a species of frog and suppose that global warming causes the components W , X , Y , and Z of its genome expression vector to change from one year to the next according to the diagram



- 19. Genetics of inbreeding** Plants are sometimes bred with themselves to generate homozygous individuals (that is,

individuals that carry two identical copies of a gene of interest). Suppose the numbers of plants that are AA , Aa , and aa in generation t are denoted by d_t , h_t , and r_t , respectively. These variables then change from one generation to the next according to the diagram



20–22 Write the system of equations and construct the corresponding matrix model for each of the following exercises.

20. DNA methylation DNA methylation is a process by which methyl chemical groups are attached to the DNA. Thus each gene in the genome can be methylated or unmethylated. Suppose that 80% of methylated genes remain methylated and 50% of unmethylated genes remain unmethylated each generation.

21. Mutation Suppose that one of three different alleles is present in each individual in a population. In each generation the following happens: 5% of individuals carrying

allele X mutate to carry allele Y , 3% mutate to allele Z , and the rest remain unchanged; 0.1% of individuals carrying allele Y mutate to carry allele Z and the rest remain unchanged; 90% of individuals carrying allele Z mutate to carry allele X and the rest remain unchanged.

22. Spotted owls were at the center of a debate between conservation biologists and the logging industry in the 1990s. Part of resolving the debate involved constructing a matrix model for their population. The owl population in year t can be divided into juveniles, subadults, and adults. Adults produce, on average, 0.33 juveniles each year. Approximately 60% of juveniles survive to be subadults the next year, while 71% of subadults survive to become adults. Approximately 94% of the adult population survives from one year to the next.



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8.6 The Inverse and Determinant of a Matrix

This section introduces the ideas of matrix inverses and determinants and illustrates how they can be used to solve systems of equations. The inverse and determinant are defined only for square matrices. Our focus in the main text is on the mathematical details, with applications explored in the exercises and subsequent sections.

■ The Inverse of a Matrix

In Section 8.4 the identity matrix I was introduced and we noted that I plays the role of the number 1 in matrix multiplication. In particular, $AI = A$ and $IA = A$ just as $a1 = a$ and $1a = a$ in scalar multiplication. In the scalar case we can rearrange the equation to obtain $a^{-1}a = 1$, where $a^{-1} = 1/a$ provided that $a \neq 0$. In other words, a^{-1} is the quantity that, when multiplied with a , gives 1. And if $a = 0$, then no such quantity exists.

Can something similar be defined for matrices? Is there a matrix that, when multiplied with A , gives the identity matrix I ? We will see that, just as with the scalar case, sometimes there is and sometimes there isn't. The existence of a matrix B such that $BA = I$ requires that the matrix A , in some sense, not “act like zero.” One of the goals of this section is to make this idea precise. When such a matrix exists, it is called the *inverse* of A .

EXAMPLE 1 | Show that B is the inverse of A , where

$$A = \begin{bmatrix} 1 & 2 \\ 7 & 5 \end{bmatrix} \quad B = \begin{bmatrix} -\frac{5}{9} & \frac{2}{9} \\ \frac{7}{9} & -\frac{1}{9} \end{bmatrix}$$

SOLUTION Calculating the matrix product gives

$$BA = \begin{bmatrix} -\frac{5}{9} & \frac{2}{9} \\ \frac{7}{9} & -\frac{1}{9} \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 7 & 5 \end{bmatrix} = \begin{bmatrix} -\frac{5}{9}(1) + \frac{2}{9}(7) & -\frac{5}{9}(2) + \frac{2}{9}(5) \\ \frac{7}{9}(1) - \frac{1}{9}(7) & \frac{7}{9}(2) - \frac{1}{9}(5) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I \quad \blacksquare$$

Definition Suppose that A is an $n \times n$ matrix. If there exists an $n \times n$ matrix B such that

$$AB = BA = I$$

then B is called the **inverse** of A and is denoted by A^{-1} .

It is possible to show that, when the inverse of a matrix exists, it is unique (see Exercise 8). It is also possible to show that, if there exists a matrix B such that $AB = I$, then necessarily $BA = I$ as well (and vice versa). Therefore we need check only one order of multiplication when finding an inverse. If A has an inverse, then we say that A is **invertible** or **nonsingular**. Otherwise A is called **singular**.

The following example illustrates how to calculate a matrix inverse in a 2×2 case, provided that the inverse exists.

EXAMPLE 2 | Derive the inverse of the matrix A from Example 1 by using the definition of an inverse.

SOLUTION From the definition, if a matrix $B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$ is the inverse of $A = \begin{bmatrix} 1 & 2 \\ 7 & 5 \end{bmatrix}$, then $AB = BA = I$. We need focus on only one of these orders of multiplication. Choosing $AB = I$, we have

$$\begin{bmatrix} 1 & 2 \\ 7 & 5 \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

or
$$\begin{bmatrix} b_{11} + 2b_{21} & b_{12} + 2b_{22} \\ 7b_{11} + 5b_{21} & 7b_{12} + 5b_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Looking at this equation entry by entry, we see that it represents four equations in four unknowns. The four equations can be split into two pairs of equations, each with two unknowns:

$$\begin{array}{ll} b_{11} + 2b_{21} = 1 & \text{and} \quad b_{12} + 2b_{22} = 0 \\ 7b_{11} + 5b_{21} = 0 & \quad \quad \quad 7b_{12} + 5b_{22} = 1 \end{array}$$

We can now solve each pair by substitution. If we focus on the first pair, we see that the second equation gives $b_{21} = -7b_{11}/5$. Substituting this into the first equation of this pair then gives $b_{11} + 2(-7b_{11}/5) = 1$, whose solution is $b_{11} = -5/9$. This can then be back-substituted into $b_{21} = -7b_{11}/5$ to give $b_{21} = 7/9$.

Similarly, the first equation of the second pair gives $b_{12} = -2b_{22}$. Substituting this into the second equation of this pair then gives $7(-2b_{22}) + 5b_{22} = 1$, whose solution is $b_{22} = -1/9$. This can then be back-substituted into $b_{12} = -2b_{22}$ to give $b_{12} = 2/9$.

Putting these results together in the matrix B gives

$$B = \begin{bmatrix} -\frac{5}{9} & \frac{2}{9} \\ \frac{7}{9} & -\frac{1}{9} \end{bmatrix} \quad \blacksquare$$

The approach used in Example 2 leads to the following formula for the inverse of a 2×2 matrix (see Exercise 4):

The Inverse of a 2×2 Matrix Suppose

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

and $a_{11}a_{22} - a_{12}a_{21} \neq 0$. Then A is invertible and

$$A^{-1} = \frac{1}{a_{11}a_{22} - a_{12}a_{21}} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}$$

If $a_{11}a_{22} - a_{12}a_{21} = 0$, then A is not invertible (that is, A is singular).

EXAMPLE 3 | If possible, find the inverse.

$$(a) \ M = \begin{bmatrix} 7 & 9 \\ 5 & 6 \end{bmatrix} \quad (b) \ N = \begin{bmatrix} 14 & 6 \\ 7 & 3 \end{bmatrix} \quad (c) \ J = \begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix}$$

SOLUTION From the formula for the inverse of a 2×2 matrix, an inverse will exist if and only if $a_{11}a_{22} - a_{12}a_{21} \neq 0$.

(a) $m_{11}m_{22} - m_{12}m_{21} = (7)(6) - (9)(5) = 42 - 45 = -3$. Therefore an inverse exists. Using the formula, we obtain

$$M^{-1} = \frac{1}{-3} \begin{bmatrix} 6 & -9 \\ -5 & 7 \end{bmatrix} = \begin{bmatrix} -2 & 3 \\ \frac{5}{3} & -\frac{7}{3} \end{bmatrix}$$

(b) $n_{11}n_{22} - n_{12}n_{21} = (14)(3) - (6)(7) = 42 - 42 = 0$. Therefore an inverse does not exist.

(c) $j_{11}j_{22} - j_{12}j_{21} = (2)(-2) - (3)(-1) = -4 + 3 = -1$. Therefore an inverse exists. Using the formula, we obtain

$$J^{-1} = \frac{1}{-1} \begin{bmatrix} -2 & -3 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix}$$

Notice that this last example has the interesting property that $J^{-1} = J$. In other words, J is its own inverse. ■

The following properties of matrix inverses are often useful. These are proved in Exercises 6, 7, and 8.

Properties of Matrix Inverses Suppose A and B are both invertible $n \times n$ matrices. Then

1. $(A^{-1})^{-1} = A$
2. $(AB)^{-1} = B^{-1}A^{-1}$
3. A^{-1} is unique.

■ The Determinant of a Matrix

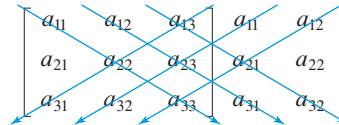
The formula for the inverse of a 2×2 matrix allows us to determine when such a matrix is invertible. But how can we tell this for matrices of other sizes? As already mentioned, a matrix A is invertible if, in some sense, it does not “act like zero.” We now make this idea precise.

To any $n \times n$ matrix A we can assign a scalar quantity called its *determinant*, denoted by $\det A$. If we view scalars as 1×1 matrices, then the definition of the determinant for matrices of sizes $n = 1$ through $n = 3$ is as follows.

The Determinant Suppose A is an $n \times n$ matrix.

1. If $n = 1$, then $\det A = a_{11}$.
2. If $n = 2$, then $\det A = a_{11}a_{22} - a_{12}a_{21}$.
3. If $n = 3$, then $\det A = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33}$.

There is an algorithmic procedure for calculating the determinant of larger matrices but it becomes impractical to do so by hand for matrices larger than 3×3 . There is also a simple graphical device for remembering the formula for the determinant of a 3×3 matrix. To begin, duplicate the first two columns to the right of the matrix. You can then compute the determinant by multiplying the entries on six diagonals, adding the rightward products and subtracting the leftward products.



Given the determinant of a matrix we have the following theorem:

(1) Theorem If A is an $n \times n$ matrix, then A is invertible if and only if $\det A \neq 0$.

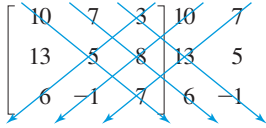
Notice that the quantity in the denominator of the formula for the inverse of a 2×2 matrix is its determinant.

EXAMPLE 4 | Which of the following matrices are invertible?

$$(a) A = [2] \quad (b) B = \begin{bmatrix} 2 & 3 \\ 6 & 9 \end{bmatrix} \quad (c) C = \begin{bmatrix} 5 & 3 \\ 2 & 1 \end{bmatrix} \quad (d) D = \begin{bmatrix} 10 & 7 & 3 \\ 13 & 5 & 8 \\ 6 & -1 & 7 \end{bmatrix}$$

SOLUTION

- The matrix A is 1×1 , with $\det A = 2$. Therefore it is invertible.
- The matrix B is 2×2 , with $\det B = (2)(9) - (3)(6) = 0$. Therefore it is not invertible.
- The matrix C is 2×2 , with $\det C = (5)(1) - (3)(2) = -1$. Therefore it is invertible.



(d) The matrix D is 3×3 , and

$$\begin{aligned}\det D &= (10)(5)(7) + (7)(8)(6) + (3)(13)(-1) - (3)(5)(6) - (10)(8)(-1) - (7)(13)(7) \\ &= 350 + 336 + (-39) - 90 - (-80) - 637 \\ &= 0\end{aligned}$$

Therefore D is not invertible. ■

■ Solving Systems of Linear Equations

One reason that matrix inverses (and therefore determinants) are useful is for solving linear systems of n equations with n unknowns. Consider, for example, the following pair of equations with two unknowns:

$$3x_1 - 2x_2 = -4$$

$$7x_1 + x_2 = 19$$

Writing these equations in matrix notation gives

$$(2) \quad \mathbf{Ax} = \mathbf{b}$$

where $A = \begin{bmatrix} 3 & -2 \\ 7 & 1 \end{bmatrix}$ is a matrix of coefficients, $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} -4 \\ 19 \end{bmatrix}$. Such a system of equations in which $\mathbf{b} \neq \mathbf{0}$ is called an **inhomogeneous** system. In general, there are three possibilities for such a system: (i) there is a unique nonzero solution for $\mathbf{x}$, (ii) there are infinitely many solutions for $\mathbf{x}$, or (iii) there is no solution for $\mathbf{x}$.

To solve Equation 2 let's first compare its form to the scalar equation.

$$(3) \quad ax = b$$

Provided that $a \neq 0$ we can solve Equation 3 by multiplying both sides by $a^{-1} = 1/a$ to get $a^{-1}ax = a^{-1}b$, or $x = a^{-1}b$. In a similar fashion, provided that $\det A \neq 0$ the matrix A will be invertible and we can multiply both sides of Equation 2 by A^{-1} to obtain

$$A^{-1}A\mathbf{x} = A^{-1}\mathbf{b}$$

$$\text{or} \quad \mathbf{x} = A^{-1}\mathbf{b}$$

Using the formula for the inverse of a 2×2 matrix, we obtain

$$\mathbf{x} = \frac{1}{(3)(1) - (-2)(7)} \begin{bmatrix} 1 & 2 \\ -7 & 3 \end{bmatrix} \begin{bmatrix} -4 \\ 19 \end{bmatrix} = \begin{bmatrix} \frac{1}{17} & \frac{2}{17} \\ -\frac{7}{17} & \frac{3}{17} \end{bmatrix} \begin{bmatrix} -4 \\ 19 \end{bmatrix} = \begin{bmatrix} \frac{-4 + 38}{17} \\ \frac{28 + 57}{17} \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

Equation 2 thus has a unique solution given by $x_1 = 2$, $x_2 = 5$. Graphically, the two equations represent straight lines in the x_1x_2 -plane and their intersection point is the unique solution illustrated in Figure 1. More generally, we have the following theorem.

(4) Theorem Suppose A is an $n \times n$ matrix, $\mathbf{x}$ is an $n \times 1$ vector of unknowns, and $\mathbf{b}$ is an $n \times 1$ vector of constants. If A is invertible, then the inhomogeneous system of equations $A\mathbf{x} = \mathbf{b}$ has a unique solution given by $\mathbf{x} = A^{-1}\mathbf{b}$.

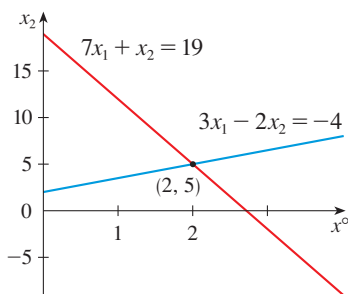


FIGURE 1

If A in Theorem 4 is not invertible, then the system of equations can have either infinitely many solutions or no solution.

EXAMPLE 5 | Using $\mathbf{b} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, obtain the solution to the inhomogeneous system of equations for each of the following matrices.

(a) $A = \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}$ (b) $B = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$ (c) $C = \begin{bmatrix} 1 & -1 \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$

SOLUTION

(a) First we calculate $\det A = 1 - (-2) = 3$ and therefore the matrix A is invertible. Theorem 4 then tells us that the inhomogeneous equation $A\mathbf{x} = \mathbf{b}$ has a unique solution given by $\mathbf{x} = A^{-1}\mathbf{b}$. Using the formula of the inverse of a 2×2 matrix, we obtain

$$\mathbf{x} = \frac{1}{3} \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} \\ -\frac{2}{3} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{2+1}{3} \\ \frac{-4+1}{3} \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

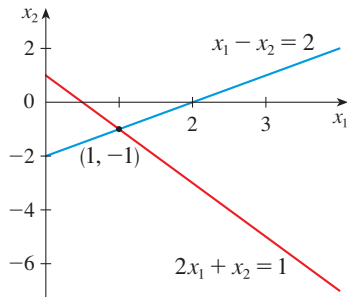


FIGURE 2

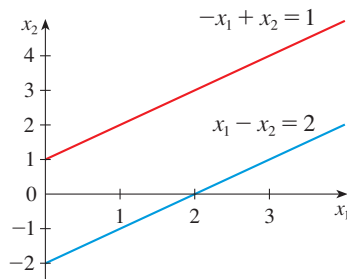


FIGURE 3

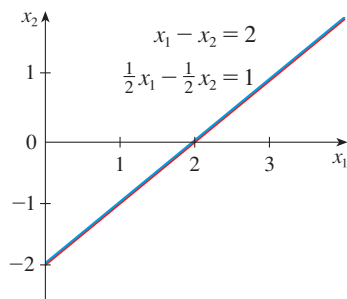


FIGURE 4

The solution corresponds to the intersection point of the lines defined by $x_1 - x_2 = 2$ and $2x_1 + x_2 = 1$, as shown in Figure 2.

(b) Calculating the determinant of B gives $\det B = 1 - 1 = 0$. Therefore the matrix B is not invertible. The inhomogeneous equation $B\mathbf{x} = \mathbf{b}$ might have an infinite number of solutions or no solution. To determine which is the case, we need to investigate the equations in more depth. Carrying out the matrix multiplication gives the two equations $x_1 - x_2 = 2$ and $-x_1 + x_2 = 1$. Solving the second equation for x_2 gives $x_2 = 1 + x_1$, and substituting this into the first equation gives $x_1 - (1 + x_1) = 2$. This simplifies to $-1 = 2$. Because there is no choice of x_1 that will make this true, there is no solution. Graphically, the two equations represent parallel straight lines in the x_1x_2 -plane and the lack of a solution corresponds to their not having an intersection point (see Figure 3).

(c) Calculating the determinant of C gives $\det C = -\frac{1}{2} - (-\frac{1}{2}) = 0$. Therefore the matrix C is not invertible and the inhomogeneous equation $C\mathbf{x} = \mathbf{b}$ might have an infinite number of solutions or no solution. Carrying out the matrix multiplication gives the two equations $x_1 - x_2 = 2$ and $\frac{1}{2}x_1 - \frac{1}{2}x_2 = 1$. Solving the second equation for x_1 gives $x_1 = 2 + x_2$, and substituting this into the first equation gives $(2 + x_2) - x_2 = 2$. This simplifies to $2 = 2$. This equation holds true no matter what value of x_2 is chosen and therefore there are an infinite number of solutions. Graphically, the two equations represent the same straight line in the x_1x_2 -plane, and therefore there are an infinite number of points of intersection as illustrated in Figure 4. ■

The solution to a system of linear equations is simplified considerably in the special case where $\mathbf{b} = \mathbf{0}$. This results in an equation of the form $A\mathbf{x} = \mathbf{0}$, which is called a **homogeneous** system. Clearly $\mathbf{x} = \mathbf{0}$ is always a solution and it is referred to as the **trivial solution**. There are therefore now only two possibilities: (i) the trivial solution is the unique solution, or (ii) there are an infinite number of nontrivial solutions. The following theorem tells us when each of these outcomes occurs.

(5) Theorem Suppose A is an $n \times n$ matrix, $\mathbf{x}$ is an $n \times 1$ vector of unknowns, and $\mathbf{0}$ is an $n \times 1$ vector of zeros. If A is invertible, then the homogeneous system of equations $A\mathbf{x} = \mathbf{0}$ has a unique solution given by the trivial solution $\mathbf{x} = \mathbf{0}$. If A is not invertible, then there are infinitely many nontrivial solutions.

EXERCISES 8.6

1. Determine if matrices A and B are inverses of one another.

$$(a) A = \begin{bmatrix} 1 & 5 \\ 2 & 7 \end{bmatrix} \quad B = \begin{bmatrix} -\frac{7}{3} & \frac{5}{3} \\ \frac{2}{3} & -\frac{1}{3} \end{bmatrix}$$

$$(b) A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 7 & 9 \\ 0 & 2 & 1 \end{bmatrix} \quad B = \begin{bmatrix} -11 & 6 & -21 \\ -2 & 1 & -3 \\ 4 & -2 & 7 \end{bmatrix}$$

$$(c) A = \begin{bmatrix} 0 & 1 \\ 3 & 2 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 3 \\ 1 & 0 \end{bmatrix}$$

$$(d) A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \quad B = \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix}$$

$$(e) A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 7 \\ 0 & 2 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 5 & 3 \\ 7 & 0 & 1 \end{bmatrix}$$

$$(f) A = \begin{bmatrix} 9 & 0 \\ 2 & 3 \end{bmatrix} \quad B = \begin{bmatrix} \frac{1}{9} & 0 \\ -\frac{2}{27} & \frac{1}{3} \end{bmatrix}$$

2. Find the inverse of the matrix.

$$(a) \begin{bmatrix} 5 & 0 \\ 9 & 6 \end{bmatrix} \quad (b) \begin{bmatrix} 1 & 3 \\ 1 & 0 \end{bmatrix}$$

$$(c) \begin{bmatrix} 9 & 2 \\ 7 & 5 \end{bmatrix} \quad (d) \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$$

$$(e) \begin{bmatrix} 3x & y \\ 2x & y \end{bmatrix} \quad (f) \begin{bmatrix} x^2 & 2x \\ x^3 & x \end{bmatrix}$$

3. Suppose that D is an $n \times n$ diagonal matrix with entries d_{ii} . Show that D^{-1} is an $n \times n$ diagonal matrix with entries $1/d_{ii}$.

4. Suppose A is an nonsingular 2×2 matrix. Derive the formula for its inverse, namely,

$$A^{-1} = \frac{1}{a_{11}a_{22} - a_{12}a_{21}} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}$$

5. Find all 2×2 matrices A such that $\det A = 1$ and $A = A^{-1}$.

6. If A is nonsingular, then $(A^{-1})^{-1} = A$.

- (a) Verify this theorem for 2×2 matrices.
(b) Prove the theorem for any $n \times n$ matrix.

7. Suppose that A and B are nonsingular matrices. Then AB is also nonsingular. Furthermore, a theorem from linear algebra then states that $(AB)^{-1} = B^{-1}A^{-1}$.

- (a) Verify this theorem for 2×2 matrices.
(b) Prove the theorem for any $n \times n$ matrix.

8. Suppose that A is nonsingular. Prove that its inverse A^{-1} is unique. [Hint: Show that if there exists two matrices, B and C , such that $BA = I$ and $AC = I$, then necessarily $B = C$.]

9. If A is nonsingular, then $(A^T)^{-1} = (A^{-1})^T$.

- (a) Verify this theorem for 2×2 matrices.
(b) Prove the theorem for any $n \times n$ matrix. [Hint: You might wish to use the property from Exercise 8.4.15(b).]

10. Use the determinant to decide whether each matrix is singular or nonsingular.

$$(a) \begin{bmatrix} 3 & 2 \\ 6 & 4 \end{bmatrix}$$

$$(b) \begin{bmatrix} a & b \\ 7a & 9b \end{bmatrix} \quad a \neq 0, b \neq 0$$

$$(c) \begin{bmatrix} 1 & 0 & 5 \\ 4 & 2 & 6 \\ 3 & 2 & 1 \end{bmatrix}$$

$$(d) \begin{bmatrix} 9 & 1 & 0 \\ 1 & 0 & 1 \\ -3 & 2 & 0 \end{bmatrix}$$

$$(e) \begin{bmatrix} x & x + x^2 \\ 3x & 0 \end{bmatrix} \quad x \neq 0, x \neq -1$$

$$(f) \begin{bmatrix} y^3 & 4y^3 \\ 3x^2 & 12x^2 \end{bmatrix}$$

11. For which values of a is the matrix singular?

$$\begin{bmatrix} a & 4 \\ 2 & 8 \end{bmatrix}$$

12. For which values of a and b is the matrix singular?

$$\begin{bmatrix} a & b \\ 7 & 9 \end{bmatrix}$$

13. For which values of a is the matrix singular?

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & a \\ 1 & 4 & a^2 \end{bmatrix}$$

14. If A and B are $n \times n$ matrices, then $\det(AB) = \det A \det B$. Verify this theorem for 2×2 matrices.
15. Using the theorem from Exercise 14, what can you say about the determinant of the matrix power A^k , where k is a positive integer, if the matrix A is nonsingular?
16. If A is an $n \times n$ matrix, then $\det A = \det A^T$. Verify this theorem for 2×2 matrices.
17. If D is an $n \times n$ diagonal matrix with entries d_{ii} , then $\det D = d_{11}d_{22} \cdots d_{nn}$. Verify this theorem for 2×2 matrices.
18. Write each system of linear equations in the form $A\mathbf{x} = \mathbf{b}$ by identifying the matrix A and the vectors $\mathbf{x}$ and $\mathbf{b}$.
- (a) $3x_1 - x_2 = 9$ (b) $2x_1 = 10$
 $2x_1 + 9x_2 = -10$ $3x_1 - x_2 = 14$
- (c) $x_1 + x_2 - x_3 = 0$ (d) $2x_1 - x_2 + 9x_3 = 12$
 $2x_1 - 2x_2 + 9x_3 = 5$ $x_1 - \frac{1}{2}x_2 + x_3 = 1$
 $x_1 + 9x_3 = 1$ $x_2 = 2$

19. For which values of k does the system of linear equations have zero, one, or an infinite number of solutions?
 [Note: Not all three possibilities need occur.]

$$\begin{aligned} 3x_1 + x_2 &= 2 \\ kx_1 + 2x_2 &= 4 \end{aligned}$$

20. For which values of k does the system of linear equations have zero, one, or an infinite number of solutions?
 [Note: Not all three possibilities need occur.]

$$\begin{aligned} 2x_1 - x_2 &= 3 \\ 4x_1 - 2x_2 &= k \end{aligned}$$

21. Consider the linear system of equations

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 &= b_1 \\ a_{21}x_1 + a_{22}x_2 &= b_2 \end{aligned}$$

Suppose that the matrix $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ is nonsingular. Derive expressions for the solutions x_1 and x_2 .

22. Suppose that $\mathbf{x} = \mathbf{p}$ and $\mathbf{x} = \mathbf{q}$ are both solutions to the inhomogeneous system of equations $A\mathbf{x} = \mathbf{b}$. Show that $\mathbf{z} = \alpha\mathbf{p} + (1 - \alpha)\mathbf{q}$ is then also a solution, where α is a scalar.
23. Suppose that $\mathbf{x} = \mathbf{p}$ and $\mathbf{x} = \mathbf{q}$ are both solutions to the homogeneous system of equations $A\mathbf{x} = \mathbf{0}$. Show that $\mathbf{z} = \alpha\mathbf{p} + \beta\mathbf{q}$ is then also a solution, where α and β are scalars.

24–29 Find all solutions to the system of linear equations.

24. $x_1 + 2x_2 = 0$ 25. $x_1 - 3x_2 = 5$
 $2x_1 + 5x_2 = 1$ $x_1 + x_2 = 2$

26. $x_1 = 1$ 27. $3x_1 + 6x_2 = 2$
 $2x_1 + 3x_2 = 6$ $9x_1 + 12x_2 = 1$
28. $5x_1 + x_2 = 0$ 29. $8x_1 + 4x_2 = 4$
 $25x_1 + 5x_2 = 0$ $4x_1 + 2x_2 = 2$

- 30. Vectorcardiography** Suppose that the voltage vector $\mathbf{v}_t$ of the heart changes from one beat to the next according to the equation

$$\mathbf{v}_{t+1} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \mathbf{v}_t$$

If the voltage vector during the current heartbeat is $\begin{bmatrix} 0.3 \\ -0.2 \end{bmatrix}$, what was the voltage vector in the previous heartbeat?

- 31. Antigenic evolution** Suppose the vector $\mathbf{x}_t$ characterizing the antigenic state of an influenza virus population changes from one season to the next according to the equation

$$\mathbf{x}_{t+1} = \begin{bmatrix} 2 & 3 \\ 0 & 0.9 \end{bmatrix} \mathbf{x}_t$$

If the vector in the current season is $\begin{bmatrix} 8 \\ 1.8 \end{bmatrix}$, what was the vector in the previous influenza season?

32. Find a second-degree polynomial (that is, an equation of the form $y = a + bx + cx^2$) that goes through the three points $(0, 1)$, $(1, 0)$, and $(-1, 0)$. Is there more than one possibility?

- 33. Leslie matrices** Consider the following model for the population size $\mathbf{n}_t$ of an age-structured population with two age classes:

$$\mathbf{n}_{t+1} = \begin{bmatrix} b & 2 \\ \frac{1}{2} & 0 \end{bmatrix} \mathbf{n}_t$$

An equilibrium is a value of the vector for which no change occurs (that is, $\mathbf{n}_{t+1} = \mathbf{n}_t$). Denoting such values by $\hat{\mathbf{n}}$, they must therefore satisfy the equation

$$\hat{\mathbf{n}} = \begin{bmatrix} b & 2 \\ \frac{1}{2} & 0 \end{bmatrix} \hat{\mathbf{n}}$$

- (a) Suppose that $b \neq 0$. Find all possible equilibrium values.
- (b) Suppose that $b = 0$. Find all possible equilibrium values.

- 34. Mutation** Suppose that, as a result of mutation, the number of individuals in a population carrying allele A and allele B changes in a way described by the equation

$$\mathbf{y}_{t+1} = \begin{bmatrix} 0.95 & 0 \\ 0.05 & 1 \end{bmatrix} \mathbf{y}_t$$

where $\mathbf{y}_t$ is the vector whose components are the numbers of individuals carrying each allele at time t . An equilibrium is a value of the vector for which no change occurs (that is, $\mathbf{y}_{t+1} = \mathbf{y}_t$). Denoting such values by $\hat{\mathbf{y}}$, they must therefore satisfy the equation

$$\hat{\mathbf{y}} = \begin{bmatrix} 0.95 & 0 \\ 0.05 & 1 \end{bmatrix} \hat{\mathbf{y}}$$

Find all possible equilibrium values.

- 35. Vectorcardiography** Suppose that the voltage vector $\mathbf{v}_t$ of the heart changes from one beat to the next according to the equation

$$\mathbf{v}_{t+1} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \mathbf{v}_t$$

An equilibrium is a value of the vector for which no change occurs (that is, $\mathbf{v}_{t+1} = \mathbf{v}_t$). Denoting such values by $\hat{\mathbf{v}}$, they

must therefore satisfy the equation

$$\hat{\mathbf{v}} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \hat{\mathbf{v}}$$

Find all possible equilibrium values.

- 36. Resource allocation** Each day an organism has 100 J of energy to divide between growth and reproduction. Each millimeter of growth costs 3 J and each egg produced costs 5 J. Denote the amount of growth per day by x_1 and the number of eggs produced per day by x_2 .
- Suppose that the organism divides its energy so that for every millimeter of growth that occurs each day, it produces (on average) two eggs. What is the amount of growth and number of eggs produced on each day?
 - Suppose that the organism divides its energy each day so that the total energy spent on eggs is twice that spent on growth. What is the amount of growth and number of eggs produced on each day?

PROJECT Cubic Splines

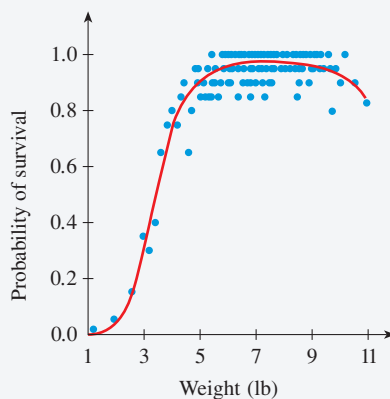


FIGURE 1

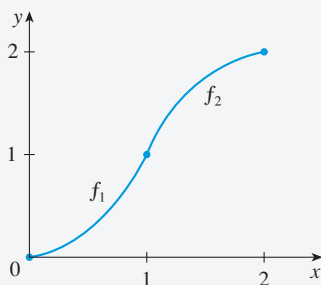


FIGURE 2

In many situations it is desirable to construct a curve that goes through specified points with certain slope and curvature properties at these points. For example, if we are interested in the relationship between birth weight and infant survival, as illustrated in Figure 1, we might wish to fit a relatively flexible curve to data.¹

One way to do this is to construct a curve that is made up of cubic polynomials, “pieced together” at specific points. Such a curve is called a *cubic spline*. We consider how to fit curves to data in more detail in Chapter 11, but for the present purposes we will simply construct a cubic spline that satisfies certain conditions.

Suppose we wish to construct a cubic spline that is made up of two segments and satisfies the following properties: (i) the spline goes through the points (0, 0), (1, 1) and (2, 2); (ii) the two segments are joined together at the point (1, 1); (iii) the slope and second derivative of the spline are both zero at the points (0, 0) and (2, 2); and (iv) the spline has continuous first and second derivatives everywhere between 0 and 2. Such a curve is shown in Figure 2.

Each segment of the cubic spline consists of a cubic polynomial of the form $f_i = a_i + b_i x + c_i x^2 + d_i x^3$.

- Consider the segment f_1 that joins the points (0, 0) and (1, 1). Obtain a system of four linear equations in four unknowns that characterizes the coefficients a_1 , b_1 , c_1 , and d_1 , and write it in matrix form.
- Solve the system of equations from Problem 1.
- Consider the segment f_2 that joins the points (1, 1) and (2, 2). Obtain a system of four linear equations in four unknowns that characterizes the coefficients a_2 , b_2 , c_2 , and d_2 , and write it in matrix form.

- CAS** 4. Solve the system of equations from Problem 3.

1. D. Schluter, “Estimating the Form of Natural Selection on a Quantitative Trait,” *Evolution* 42 (1988): 849–61.

For Problem 2 you should be able to solve the system of equations by inspection. For Problem 4, however, to solve the system of equations using matrix inverses you will need to calculate the inverse of a 4×4 matrix by using a computer algebra system.

8.7 Eigenvectors and Eigenvalues

The mathematical background of Section 8.6 can now be used to return to our study of matrix models. In this section we introduce eigenvectors and eigenvalues. In Section 8.8 we then show how they are fundamental to understanding the behavior of matrix models.

■ Characterizing How Matrix Multiplication Changes Vectors

In Section 8.5 we studied models for the dynamics of vectors having the form

$$\mathbf{n}_{t+1} = A\mathbf{n}_t$$

where $\mathbf{n}_t$ is a vector of variables and A is a square matrix. Our goal now is to characterize in general what such multiplication of a vector by a matrix does to the vector.

As an example, suppose that

$$A = \begin{bmatrix} 1 & \frac{1}{2} \\ 1 & \frac{3}{2} \end{bmatrix}$$

Figure 1 shows how this matrix changes the vector from one time step to the next when $\mathbf{n}_0 = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$. The changes in the vector look quite complicated. For example, the direction of the vector keeps changing and sometimes the vector is compressed from one step to the next (for example, between time 0 and time 1) and sometimes it is stretched (for example, between time 2 and time 3).

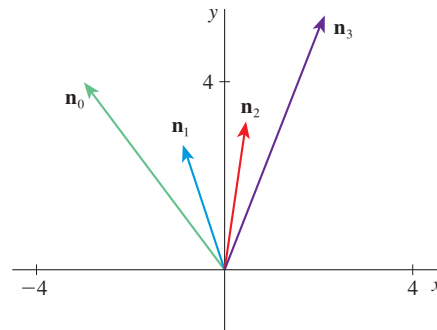


FIGURE 1

If you were to experiment with different initial vectors $\mathbf{n}_0$, you would get different patterns. However, you might happen across an initial vector that results in a particularly simple result. For example, suppose $\mathbf{n}_0 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. In this case the vector remains on its

initial axis, and it is simply stretched by a factor of 2 each time step, as shown in Figure 2(a). Likewise, if $\mathbf{n}_0 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ then again the vector remains on its initial axis, but now it is compressed by a factor of $\frac{1}{2}$ each time step, as shown in Figure 2(b). Thus, for some special initial vectors, the change in the vector is relatively easy to describe.

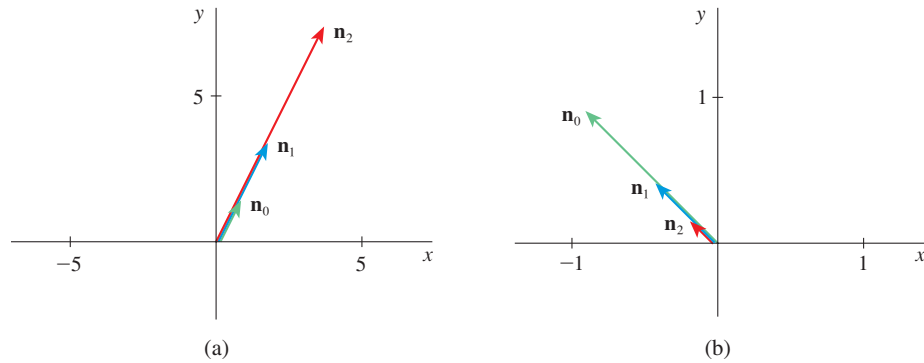


FIGURE 2

As we will see in Section 8.8, such special vectors play a central role in understanding the behavior of matrix models. These vectors are called *eigenvectors* of the matrix A , and the factor by which each vector is stretched or shrunk is called its corresponding *eigenvalue*. In the remainder of this section we will learn how to calculate eigenvectors and eigenvalues. We will see that a matrix of size n has exactly n eigenvalues, although our focus will be primarily on $n = 2$. We also focus on matrices whose eigenvalues are distinct (that is, none of them have the same value). This is a sufficient condition for each eigenvalue to be associated with its own distinct eigenvector. Matrices for which this is not true are called **defective** and are studied in courses on linear algebra.

■ Eigenvectors and Eigenvalues

An eigenvector of a matrix is a vector that, when multiplied by the matrix, is simply changed in length.

(1) Definition Suppose that A is an $n \times n$ matrix. A *nonzero* vector $\mathbf{v}$ that satisfies the equation

$$A\mathbf{v} = \lambda\mathbf{v}$$

is called an **eigenvector** of the matrix A . The scalar λ is the **eigenvalue** associated with this eigenvector.

Definition 1 provides a precise description of the idea that eigenvectors always remain on their initial axis when multiplied by a matrix. If $\mathbf{v}$ is an eigenvector of A , then when we multiply $\mathbf{v}$ by A we obtain the vector $\mathbf{v}$ multiplied by a scalar λ .

How do we find eigenvectors and eigenvalues? As the following calculations reveal, it is easiest to first calculate the eigenvalues of a matrix and then calculate their associated eigenvectors.

Consider the matrix $A = \begin{bmatrix} 1 & \frac{1}{2} \\ 1 & \frac{3}{2} \end{bmatrix}$ from our earlier example. From Definition (1) we have

$$A\mathbf{v} = \lambda\mathbf{v}$$

or, equivalently,

$$A\mathbf{v} - \lambda\mathbf{v} = \mathbf{0}$$

where $\mathbf{0}$ is the zero vector. To proceed further, we factor out the vector $\mathbf{v}$. To do so, we must first multiply λ by the identity matrix I in order for the elements of the equation to maintain compatible sizes:

$$(2) \quad (A - \lambda I)\mathbf{v} = \mathbf{0}$$

Notice that, although λ is a scalar, $A - \lambda I$ is a 2×2 matrix.

We now seek a vector $\mathbf{v}$ that satisfies Equation 2. Recalling Theorem 8.6.5, we see that, if the matrix $A - \lambda I$ is invertible, then the only solution is the trivial solution $\mathbf{v} = \mathbf{0}$. Therefore, to have a nonzero eigenvector $\mathbf{v}$, we require that $A - \lambda I$ be singular. From Theorem 8.6.1, this requires that

$$\det(A - \lambda I) = 0$$

Our considerations have allowed us to remove $\mathbf{v}$ from the equation, and therefore we now have an equation that determines the eigenvalues λ . Calculating the matrix $A - \lambda I$, we get

$$A - \lambda I = \begin{bmatrix} 1 - \lambda & \frac{1}{2} \\ 1 & \frac{3}{2} - \lambda \end{bmatrix}$$

and, using the definition of the determinant of a 2×2 matrix, we have

$$\begin{aligned} \det \begin{bmatrix} 1 - \lambda & \frac{1}{2} \\ 1 & \frac{3}{2} - \lambda \end{bmatrix} &= (1 - \lambda)\left(\frac{3}{2} - \lambda\right) - \frac{1}{2} \cdot 1 \\ &= \frac{3}{2} - \lambda - \frac{3}{2}\lambda + \lambda^2 - \frac{1}{2} \\ &= \lambda^2 - \frac{5}{2}\lambda + 1 \end{aligned}$$

Setting this result equal to zero and multiplying the equation by 2 then gives $2\lambda^2 - 5\lambda + 2 = 0$. This can be factored to produce $(\lambda - 2)(2\lambda - 1) = 0$, the solutions of which are $\lambda = 2$ and $\lambda = \frac{1}{2}$. These are the eigenvalues of A .

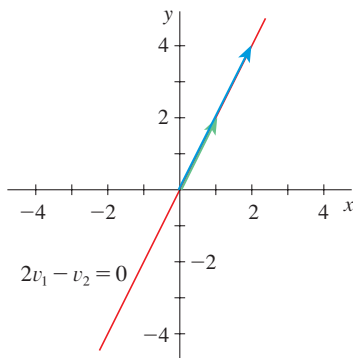
We can now calculate the eigenvector associated with each of the eigenvalues $\lambda = 2$ and $\lambda = \frac{1}{2}$. First let's find the eigenvector associated with $\lambda = 2$. We seek a vector $\mathbf{v}$ such that, when $\lambda = 2$ is substituted into the left side of Equation 2, we get the zero vector. In other words,

$$\begin{bmatrix} 1 - \lambda & \frac{1}{2} \\ 1 & \frac{3}{2} - \lambda \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 1 - 2 & \frac{1}{2} \\ 1 & \frac{3}{2} - 2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

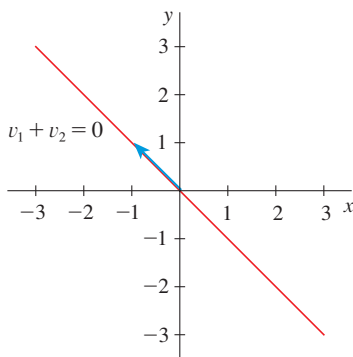
This gives the following pair of equations in two unknowns:

$$\begin{aligned} -v_1 + \frac{1}{2}v_2 &= 0 \\ v_1 - \frac{1}{2}v_2 &= 0 \end{aligned}$$

These two equations are redundant because both specify that $2v_1 = v_2$. As a result, there are infinitely many solutions—we are free to choose either v_1 or v_2 arbitrarily, and the other is then determined by this choice.

**FIGURE 3**

All eigenvectors associated with the eigenvalue $\lambda = 2$ lie on the red line.

**FIGURE 4**

All eigenvectors associated with the eigenvalue $\lambda = \frac{1}{2}$ lie on the red line.

To make our calculations simple, it is usually best to work with whole numbers. For example, we might choose $v_1 = 1$, in which case we then have $v_2 = 2$. The eigenvector associated with eigenvalue $\lambda = 2$ is then $\mathbf{v} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. If we had made a different choice for v_1 , we would have ended up with a different vector $\mathbf{v}$. For example, if we choose $v_1 = 2$, we would then have $v_2 = 4$ and thus $\mathbf{v} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$. Regardless of our choice, however, all the resulting vectors are eigenvectors and all lie on the same line because they all have the form $\mathbf{v} = \begin{bmatrix} a \\ 2a \end{bmatrix}$ or $\mathbf{v} = a \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ for some scalar a as shown in Figure 3.

We now calculate the eigenvector associated with $\lambda = \frac{1}{2}$. We obtain

$$\begin{bmatrix} 1 - \lambda & \frac{1}{2} \\ 1 & \frac{3}{2} - \lambda \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 1 - \frac{1}{2} & \frac{1}{2} \\ 1 & \frac{3}{2} - \frac{1}{2} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

which gives the pair of equations

$$\begin{aligned} \frac{1}{2}v_1 + \frac{1}{2}v_2 &= 0 \\ v_1 + v_2 &= 0 \end{aligned}$$

Again, these equations are redundant, both specifying that $v_1 = -v_2$. Choosing $v_2 = 1$, we then have $v_1 = -1$ and therefore $\mathbf{v} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$. Again the choice $v_2 = 1$ is arbitrary, but all choices result in vectors lying on the same line because they all have the form $\mathbf{v} = a \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ for some scalar a . (See Figure 4.)

Note that the eigenvector-eigenvalue pairs we just obtained are exactly those identified in Figure 2.

EXAMPLE 1 | Consider the matrix $A = \begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix}$.

- Find its eigenvalues.
- Find the eigenvectors associated with the eigenvalues from part (a).

SOLUTION

- Using the equation $\det(A - \lambda I) = 0$, we get

$$\det \begin{bmatrix} 2 - \lambda & 2 \\ 1 & 3 - \lambda \end{bmatrix} = 0$$

$$\text{or} \quad (2 - \lambda)(3 - \lambda) - 2 = \lambda^2 - 5\lambda + 4 = (\lambda - 1)(\lambda - 4) = 0$$

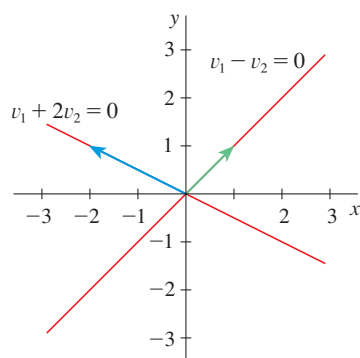
The eigenvalues are therefore $\lambda = 1$ and $\lambda = 4$.

- Beginning with eigenvalue $\lambda = 1$, we can use Equation 2 to obtain

$$\begin{bmatrix} 2 - \lambda & 2 \\ 1 & 3 - \lambda \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 2 - 1 & 2 \\ 1 & 3 - 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This is the pair of equations

$$\begin{aligned} v_1 + 2v_2 &= 0 \\ v_1 + 2v_2 &= 0 \end{aligned}$$

**FIGURE 5**

The eigenvectors for both eigenvalues, along with red lines indicating all scalar multiples of these eigenvectors.

These are identical, both specifying that $v_1 = -2v_2$. We choose (arbitrarily) $v_2 = 1$, giving $v_1 = -2$ and thus $\mathbf{v} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$.

For the eigenvalue $\lambda = 4$, we use Equation 2 to obtain

$$\begin{bmatrix} 2 - \lambda & 2 \\ 1 & 3 - \lambda \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 2 - 4 & 2 \\ 1 & 3 - 4 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

or, equivalently,

$$\begin{aligned} -2v_1 + 2v_2 &= 0 \\ v_1 - v_2 &= 0 \end{aligned}$$

Both equations specify that $v_1 = v_2$ and we choose (arbitrarily) $v_1 = 1$, giving $v_2 = 1$ and thus $\mathbf{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. (See Figure 5.)

EXAMPLE 2 | Consider the matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & \frac{1}{2} \end{bmatrix}$.

- Find its eigenvalues.
- Find the eigenvectors associated with the eigenvalues from part (a).

SOLUTION

- Using the equation $\det(A - \lambda I) = 0$, we get

$$\det \begin{bmatrix} 2 - \lambda & 1 \\ 1 & \frac{1}{2} - \lambda \end{bmatrix} = 0$$

$$\text{or} \quad (2 - \lambda)\left(\frac{1}{2} - \lambda\right) - 1 = \lambda\left(\lambda - \frac{5}{2}\right) = 0$$

The eigenvalues are therefore $\lambda = 0$ and $\lambda = \frac{5}{2}$.

- Beginning with eigenvalue $\lambda = 0$, we have

$$\begin{bmatrix} 2 - \lambda & 1 \\ 1 & \frac{1}{2} - \lambda \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 2 - 0 & 1 \\ 1 & \frac{1}{2} - 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

or, equivalently,

$$\begin{aligned} 2v_1 + v_2 &= 0 \\ v_1 + \frac{1}{2}v_2 &= 0 \end{aligned}$$

Both equations specify that $2v_1 = -v_2$. We choose $v_1 = 1$, giving the eigenvector

$$\mathbf{v} = \begin{bmatrix} 1 \\ -2 \end{bmatrix} \text{ associated with the eigenvalue } \lambda = 0.$$

For the eigenvalue $\lambda = \frac{5}{2}$, we have

$$\begin{bmatrix} 2 - \lambda & 1 \\ 1 & \frac{1}{2} - \lambda \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 2 - \frac{5}{2} & 1 \\ 1 & \frac{1}{2} - \frac{5}{2} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

or, equivalently,

$$\begin{aligned} -\frac{1}{2}v_1 + v_2 &= 0 \\ v_1 - 2v_2 &= 0 \end{aligned}$$

These equations both specify that $v_1 = 2v_2$. We choose $v_2 = 1$, giving the eigenvector

$\mathbf{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ associated with the eigenvalue $\lambda = \frac{5}{2}$. (See Figure 6).

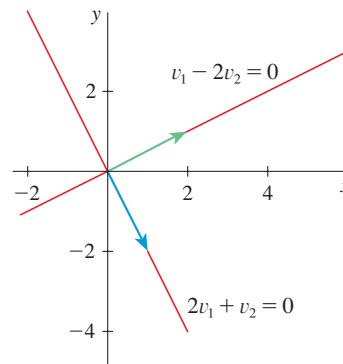


FIGURE 6

The eigenvectors for both eigenvalues, along with red lines indicating all scalar multiples of these eigenvectors.

This example illustrates that a matrix can have an eigenvalue that is zero. In Example 8.8.2, we will see what happens geometrically when vectors are multiplied by such matrices.

It is possible to derive an explicit formula for the eigenvalues of an arbitrary 2×2 matrix. Consider the matrix

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

Again, using the equation $\det(A - \lambda I) = 0$, we get

$$\det \begin{bmatrix} a_{11} - \lambda & a_{12} \\ a_{21} & a_{22} - \lambda \end{bmatrix} = 0$$

or

$$(a_{11} - \lambda)(a_{22} - \lambda) - a_{12}a_{21} = \lambda^2 - (a_{11} + a_{22})\lambda + (a_{11}a_{22} - a_{12}a_{21}) = 0$$

The last equation is a quadratic polynomial in λ , and it can therefore be solved using the quadratic formula (or perhaps by factoring).

In the preceding calculation we saw that the equation $\det(A - \lambda I) = 0$ is a second-degree polynomial in λ . Although we won't prove it, for a square matrix A of size n , the equation $\det(A - \lambda I) = 0$ that determines its eigenvalues is an n th-degree polynomial in λ . This polynomial is referred to as the **characteristic polynomial** of the matrix A . Our focus on matrices that have distinct eigenvalues is therefore a focus on matrices whose characteristic polynomial has no repeated roots.

Given that the eigenvalues of a matrix are the roots of a polynomial, we also must expect that eigenvalues are sometimes complex numbers. In general, if we define the quantity $i = \sqrt{-1}$, the eigenvalues of matrices whose entries are real numbers always come in complex conjugate pairs, having the form $\lambda = a + bi$ and $\lambda = a - bi$ for some real numbers a and b . (See Exercise 8.8.27.)

Appendix G provides a review of complex numbers.

EXAMPLE 3 | Consider the matrix $A = \begin{bmatrix} \sqrt{3} & -1 \\ 1 & \sqrt{3} \end{bmatrix}$.

- Find its eigenvalues.
- Find the eigenvectors associated with the eigenvalues from part (a).

- (c) Consider an arbitrary initial vector $\mathbf{n}_0 = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$. How does multiplication of $\mathbf{n}_0$ by A affect the length of this vector?
- (d) Use the dot product to determine how multiplication by A affects the direction of the vector.
- (e) Describe, overall, what multiplication by the matrix A does to vectors.

SOLUTION

(a) Calculating the characteristic polynomial and solving it using the quadratic formula gives the eigenvalues:

$$\begin{aligned}\lambda &= \frac{1}{2} \left[2\sqrt{3} \pm \sqrt{(-2\sqrt{3})^2 - 4(4)} \right] \\ &= \frac{1}{2} (2\sqrt{3} \pm \sqrt{12 - 16}) \\ &= \sqrt{3} \pm i\end{aligned}$$

(b) Beginning with eigenvalue $\lambda = \sqrt{3} + i$, we have

$$\begin{bmatrix} \sqrt{3} - \lambda & -1 \\ 1 & \sqrt{3} - \lambda \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} -i & -1 \\ 1 & -i \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Both of these equations specify that $v_2 = -iv_1$. We choose $v_1 = 1$, giving $\mathbf{v} = \begin{bmatrix} 1 \\ -i \end{bmatrix}$.

For eigenvalue $\lambda = \sqrt{3} - i$, we have

$$\begin{bmatrix} \sqrt{3} - \lambda & -1 \\ 1 & \sqrt{3} - \lambda \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} i & -1 \\ 1 & i \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

These equations both specify that $v_2 = iv_1$. We choose $v_1 = 1$, giving $\mathbf{v} = \begin{bmatrix} 1 \\ i \end{bmatrix}$. Notice that, when the eigenvalues are complex, their corresponding eigenvectors are also complex.

(c) Carrying out the matrix multiplication with the initial vector $\mathbf{n}_0 = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$ gives

$$\mathbf{n}_1 = \begin{bmatrix} \sqrt{3} & -1 \\ 1 & \sqrt{3} \end{bmatrix} \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} = \begin{bmatrix} \sqrt{3}x_0 - y_0 \\ x_0 + \sqrt{3}y_0 \end{bmatrix}$$

The length of the initial vector is $|\mathbf{n}_0| = \sqrt{x_0^2 + y_0^2}$, and the length of the vector $\mathbf{n}_1$ is

$$\begin{aligned}|\mathbf{n}_1| &= \sqrt{(\sqrt{3}x_0 - y_0)^2 + (x_0 + \sqrt{3}y_0)^2} \\ &= \sqrt{3x_0^2 - 2\sqrt{3}x_0y_0 + y_0^2 + x_0^2 + 2\sqrt{3}x_0y_0 + 3y_0^2} \\ &= 2\sqrt{x_0^2 + y_0^2}\end{aligned}$$

Therefore multiplication by A increases the length of the vector by a factor of 2.

(d) From the definition of the dot product, we have

$$\cos \theta = \frac{\mathbf{n}_0 \cdot \mathbf{n}_1}{|\mathbf{n}_0||\mathbf{n}_1|}$$

where θ is the angle between $\mathbf{n}_0$ and $\mathbf{n}_1$. Substituting the vectors $\mathbf{n}_0$ and $\mathbf{n}_1$ gives

Here we are writing $\mathbf{n}_0$ and $\mathbf{n}_1$ as row vectors.

$$\begin{aligned}\cos \theta &= \frac{[x_0, y_0] \cdot [\sqrt{3}x_0 - y_0, x_0 + \sqrt{3}y_0]}{2\sqrt{x_0^2 + y_0^2}\sqrt{x_0^2 + y_0^2}} \\ &= \frac{\sqrt{3}x_0^2 - x_0y_0 + x_0y_0 + \sqrt{3}y_0^2}{2(x_0^2 + y_0^2)} \\ &= \frac{\sqrt{3}}{2}\end{aligned}$$

Solving for θ shows that the angle between $\mathbf{n}_0$ and $\mathbf{n}_1$ is $\theta = \pi/6$. Thus multiplication by A rotates the vector by 30 degrees.

(e) From parts (c) and (d) and from Figure 7, we see that each multiplication by A rotates the vector 30 degrees in the counterclockwise direction and stretches its length by a factor of 2.

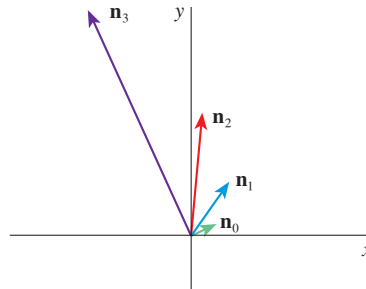
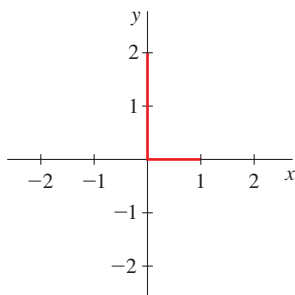


FIGURE 7

Example 3 suggests that when the eigenvalues are complex, there are no longer vectors in the x_1x_2 -plane that remain on a fixed axis when multiplied by the matrix. (There are still such vectors, but they do not lie in the x_1x_2 -plane because they are complex.) Instead, it appears that all vectors in the x_1x_2 -plane get rotated through a fixed angle. As the next section will show, this is true in general when eigenvalues are complex.

EXERCISES 8.7

1–4 Multiplication by a matrix has an interesting geometric interpretation. Consider the letter L in the plane, made up of the two vectors $[0, 2]$ and $[1, 0]$:



Describe how multiplication by the given matrix changes L.

1. The following matrices are sometimes called *reflections*.

(a) $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

(b) $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

(d) $\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$

2. The following matrices are sometimes called *contractions* or *expansions*.

(a) $\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{3} \end{bmatrix}$

(c) $\begin{bmatrix} \frac{1}{5} & 0 \\ 0 & 2 \end{bmatrix}$

(d) $\begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$

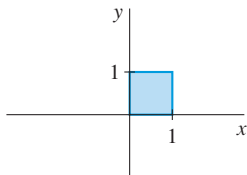
3. The following matrices are sometimes called *shears*.

$$\begin{array}{ll} \text{(a)} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} & \text{(b)} \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \\ \text{(c)} \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} & \text{(d)} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \end{array}$$

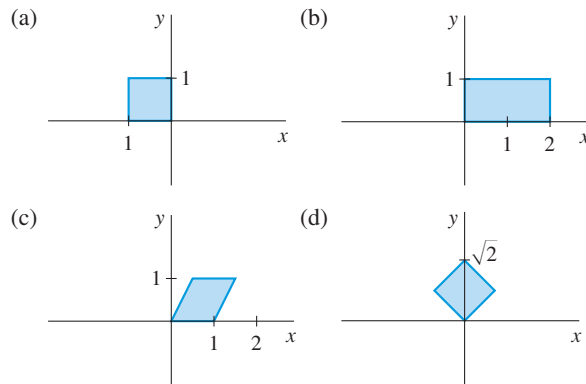
4. The following matrices are sometimes called *rotations*.

$$\begin{array}{ll} \text{(a)} \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} & \text{(b)} \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} \end{array}$$

5. Consider a unit square in the first quadrant.



Each point in the square can be viewed as the tip of its position vector. What is the matrix that results in each of the following images when applied to these vectors?



6. Use the dot product to show that the matrix

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

rotates vectors through an angle θ . Choose a particular value of θ and convince yourself that the direction of rotation is counterclockwise.

7. **Vectorcardiography** Suppose the voltage vector of a healthy heart is given by $[0.3, -0.2]$. Cardiac abnormalities result in changes to this vector. Each abnormality can be viewed as a change of a healthy heart voltage vector into a pathological voltage vector through matrix multiplication. For each of the following heart pathologies, find a matrix that produces the pathological voltage vector from a healthy voltage vector.

- (a) Left anterior hemiblock: $\mathbf{h} = [0.3, 0.2]$

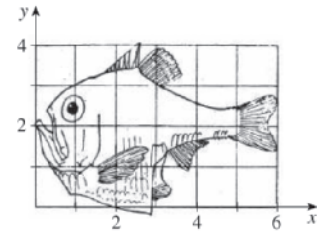
- (b) Left posterior hemiblock: $\mathbf{h} = [-0.3, -0.2]$

- (c) Apical ischemia: $\mathbf{h} = [-0.3, 0.2]$

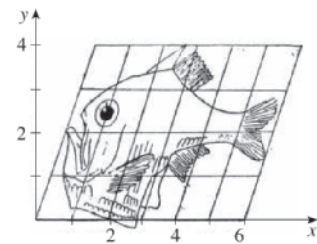
- (d) Chronic obstructive pulmonary disease:

$$\mathbf{h} = [0.1, -0.0667]$$

8. **Morphometrics** D'arcy Thompson was a Scottish scientist who pioneered the use of mathematics for describing differences in morphology between species. For example, he demonstrated how the shape of the fish species *Argyropelecus offers* can be related to the shape of another species *Sternoptyx diaphana* through matrix multiplication:



Argyropelecus offers



Sternoptyx diaphana

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What is the form of the matrix that describes this change?

9. Determine whether or not the given scalar k is an eigenvalue of the matrix A .

(a) $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ $k = 3$

(b) $A = \begin{bmatrix} 0 & 2 & 1 \\ 2 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix}$ $k = 0$

(c) $A = \begin{bmatrix} 5 & 2 \\ 0 & 1 \end{bmatrix}$ $k = 2$

(d) $A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$ $k = 1 - i$

(e) $A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & 0 \\ 2 & 1 & 0 \end{bmatrix}$ $k = 0$

(f) $A = \begin{bmatrix} 1 & a \\ 1 & 1 \end{bmatrix}$ $k = 1 + \sqrt{a}$

10. Determine whether or not $\mathbf{x}$ is an eigenvector of A . If it is, determine its associated eigenvalue.

(a) $A = \begin{bmatrix} 3 & -1 \\ 2 & 0 \end{bmatrix}$ $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

(b) $A = \begin{bmatrix} -3 & -1 & 5 \\ -2 & 1 & 2 \\ -2 & -1 & 4 \end{bmatrix}$ $\mathbf{x} = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$

(c) $A = \begin{bmatrix} 1 & -1 \\ -2 & 0 \end{bmatrix}$ $\mathbf{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

(d) $A = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}$ $\mathbf{x} = \begin{bmatrix} 2 \\ 2i \end{bmatrix}$

(e) $A = \begin{bmatrix} 2 + 3a & -2 - 2a \\ 3 + 3a & -3 - 2a \end{bmatrix}$ $\mathbf{x} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

(f) $A = \begin{bmatrix} -9 & 4 & 6 \\ -6 & 3 & 4 \\ -9 & 4 & 6 \end{bmatrix}$ $\mathbf{x} = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}$

11. Find the eigenvalues of each matrix.

(a) $\begin{bmatrix} 2 & 0 \\ 3 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 5 & -4 \\ 6 & -5 \end{bmatrix}$

(c) $\begin{bmatrix} 3 & -1 \\ 0 & 2 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 2 \\ -\frac{1}{2} & 0 \end{bmatrix}$

(e) $\begin{bmatrix} 6 & -4 & -4 \\ 0 & 0 & 0 \\ 6 & -4 & -4 \end{bmatrix}$ (f) $\begin{bmatrix} -1 & 4 & 2 \\ 0 & 3 & 0 \\ -3 & 4 & 4 \end{bmatrix}$

12. Find an eigenvector associated with the given eigenvalue of A .

(a) $A = \begin{bmatrix} 9 & 0 \\ 2 & 3 \end{bmatrix}$ $\lambda = 9$

(b) $A = \begin{bmatrix} 1 & 5 \\ 2 & 7 \end{bmatrix}$ $\lambda = 4 + \sqrt{19}$

(c) $A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 0 & 0 \\ 0 & 2 & 1 \end{bmatrix}$ $\lambda = 3$

(d) $A = \begin{bmatrix} 0 & 1 \\ 3 & 2 \end{bmatrix}$ $\lambda = -1$

(e) $A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$ $\lambda = \frac{1 + \sqrt{5}}{2}$

(f) $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 7 \\ 0 & 2 & 1 \end{bmatrix}$ $\lambda = 1 + \sqrt{14}$

13. Find the eigenvalues and eigenvectors of the matrix.

(a) $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 2 & 7 \\ 0 & 5 \end{bmatrix}$

(e) $\begin{bmatrix} 1 & 2 \\ 3 & -3 \end{bmatrix}$ (f) $\begin{bmatrix} 2 & 6 \\ 5 & 0 \end{bmatrix}$

(g) $\begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}$ (h) $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 7 \\ 0 & 2 & 1 \end{bmatrix}$

14. Derive a general formula for the eigenvalues of the 2×2 matrix

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

15. The case of matrices with *repeated eigenvalues* is treated in courses on linear algebra. As an example, try calculating the eigenvalues and eigenvectors of the following matrices. Comment on anything unusual that occurs.

(a) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

16. In general, the eigenvalues of a diagonal matrix are given by the entries on the diagonal. Verify this for 2×2 and 3×3 matrices.

17. In general, the eigenvalues of an upper triangular matrix are given by the entries on the diagonal. The same is true for a lower triangular matrix. Verify this for 2×2 and 3×3 matrices.

18. The *trace* of a matrix is given by the sum of its diagonal entries. In general, the trace of a matrix is equal to the sum of its eigenvalues, and its determinant is equal to the product of its eigenvalues. Verify this for 2×2 matrices.

19. Suppose that $A^2 = 0$ for some matrix A . Prove that the only possible eigenvalues of A are then 0.

20. Suppose that an eigenvalue of matrix A is zero. Prove that A must therefore be singular.

21. Show that the characteristic polynomials of A and A^T are the same.

22. Suppose that A is a nonsingular matrix with an eigenvalue λ . Show that $1/\lambda$ is then an eigenvalue of A^{-1} .

23. Suppose that λ is an eigenvalue of A . Show that 2λ is then an eigenvalue of $2A$.

24. Suppose that λ is an eigenvalue of A . Show that λ^2 is then an eigenvalue of A^2 .
25. Suppose that $\mathbf{v}$ is an eigenvector of matrix A with eigenvalue λ_A , and it is also an eigenvector of matrix B with eigenvalue λ_B .
- (a) Show that $\mathbf{v}$ is an eigenvector of $A + B$ and find its associated eigenvalue.
- (b) Show that $\mathbf{v}$ is an eigenvector of AB and find its associated eigenvalue.
26. **Vectorcardiography** The voltage vector of a heart with a certain pathology changes from one beat to the next as described by the matrix

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Find its eigenvalues and associated eigenvectors.

27. The **antigenic evolution** of a virus in one season is described by the matrix

$$\begin{bmatrix} 2 & 3 \\ 0 & \frac{9}{10} \end{bmatrix}$$

Find its eigenvalues and associated eigenvectors.

28. The change in **population size** of a species with juvenile and adult individuals is described by the matrix

$$\begin{bmatrix} 0 & 2 \\ \frac{1}{2} & \frac{1}{3} \end{bmatrix}$$

Find its eigenvalues and associated eigenvectors.

29. The **Leslie matrix** for an age-structured population is given by

$$\begin{bmatrix} b & 2 \\ \frac{1}{2} & 0 \end{bmatrix}$$

Find its eigenvalues and associated eigenvectors.

- CAS** 30. **Cancer progression** Example 8.5.2 introduced a model for the progression of cancer that involved the matrix

$$\begin{bmatrix} 0.998 & 0.45 & 0 \\ 0.002 & 0.1 & 0 \\ 0 & 0.45 & 1 \end{bmatrix}$$

Find its eigenvalues and associated eigenvectors.

31. The **Leslie matrix** for an age-structured population is given by

$$\begin{bmatrix} 1 & 2 & 4 \\ \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{3} & 0 \end{bmatrix}$$

Find its characteristic polynomial.

32. **The genetics of inbreeding** A model for the genetics of inbreeding was introduced in Exercise 8.5.19 and can be described by the matrix

$$\begin{bmatrix} 1 & \frac{1}{4} & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & \frac{1}{4} & 1 \end{bmatrix}$$

Try calculating its eigenvalues and associated eigenvectors. You will find that one of the eigenvalues is repeated.

8.8 Iterated Matrix Models

Matrix models are used in the life sciences to model how vectors change over multiple time steps. Section 8.5 introduced several examples. One way to determine the long-term behavior of such vectors over time is to iterate the matrix multiplication repeatedly. Given an initial vector $\mathbf{n}_0$, we calculate successive vectors $\mathbf{n}_1, \mathbf{n}_2, \dots$ using the recursion

$$(1) \quad \mathbf{n}_{t+1} = A\mathbf{n}_t$$

We used this idea in Chapters 1 and 2 to explore the scalar recursion equation $n_{t+1} = Rn_t$. As in the scalar case, however, this becomes cumbersome for large values of t . A better alternative is to find the solution to the recursion.

■ Solving Matrix Models

To solve the recursion equation (1), let's begin by repeatedly iterating it, starting with initial vector $\mathbf{n}_0$. For $\mathbf{n}_1$ we obtain

$$\mathbf{n}_1 = A\mathbf{n}_0$$

Likewise, for $\mathbf{n}_2$ we get

$$\mathbf{n}_2 = A\mathbf{n}_1 = A(A\mathbf{n}_0) = A^2\mathbf{n}_0$$

where A^2 is A multiplied with itself. Continuing, we see the general pattern

$$(2) \quad \mathbf{n}_t = A^t \mathbf{n}_0$$

where A^t is A multiplied with itself t times.

Equation 2 is the solution to recursion equation (1), just as $n_t = R^t n_0$ is the solution to the scalar recursion equation $n_{t+1} = R n_t$. To verify this we substitute $A^t \mathbf{n}_0$ into the left side of Equation 1:

$$A^{t+1} \mathbf{n}_0 = A A^t \mathbf{n}_0 = A \mathbf{n}_t$$

This is identical to the right side of Equation 1, demonstrating that it is a solution.

Unlike the scalar case, however, the matrix multiplication required to calculate A^t becomes very tedious, particularly if there are unspecified constants in A (try calculating A^3 for an arbitrary 2×2 matrix). Furthermore, such calculations do little to help us understand the long-term behavior of $\mathbf{n}_t$. This is where the eigenvalues and eigenvectors of A enter the picture. We can use them to rewrite A in a more useful way.

Recall the definition of the eigenvectors and eigenvalues of A :

$$(3) \quad A \mathbf{v}_i = \lambda_i \mathbf{v}_i$$

Note that we have now used the subscript i on $\mathbf{v}$ and λ to reflect the fact that there are n such pairs for an $n \times n$ matrix (ignoring *defective* matrices; see Section 8.7). For example, in the 2×2 case we have two eigenvectors, which we will denote as column

vectors by $\begin{bmatrix} | \\ \mathbf{v}_1 \\ | \end{bmatrix}$ and $\begin{bmatrix} | \\ \mathbf{v}_2 \\ | \end{bmatrix}$. These eigenvectors are associated with eigenvalues λ_1 and λ_2 , respectively. We can write Equation 3 simultaneously for both pairs as

$$A \begin{bmatrix} | & | \\ \mathbf{v}_1 & \mathbf{v}_2 \\ | & | \end{bmatrix} = \begin{bmatrix} | & | \\ \mathbf{v}_1 & \mathbf{v}_2 \\ | & | \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$$

where $\begin{bmatrix} | & | \\ \mathbf{v}_1 & \mathbf{v}_2 \\ | & | \end{bmatrix}$ is a 2×2 matrix whose columns are the eigenvectors $\begin{bmatrix} | \\ \mathbf{v}_1 \\ | \end{bmatrix}$ and $\begin{bmatrix} | \\ \mathbf{v}_2 \\ | \end{bmatrix}$.

More generally, for an $n \times n$ matrix A we can define an $n \times n$ matrix P whose columns are the n eigenvectors of A , and an $n \times n$ diagonal matrix D whose entries along the diagonal are the corresponding eigenvalues. We can then write

$$AP = PD$$

Our goal now is to use this equation to rewrite A in a different form, using its eigenvectors and eigenvalues. A theorem from linear algebra states that if A is not defective, then P is invertible. Therefore, multiplying both sides of the equation on the right by P^{-1} gives

$$(4) \quad A = PDP^{-1}$$

How does the expression for A in Equation 4 help us evaluate $A^t \mathbf{n}_0$ in Equation 2? Let's iterate Equation 1 again, this time using our new way of expressing A . For $\mathbf{n}_1$ we obtain

$$\mathbf{n}_1 = A \mathbf{n}_0 = PDP^{-1} \mathbf{n}_0$$

For $\mathbf{n}_2$ we get

$$\begin{aligned}\mathbf{n}_2 &= A\mathbf{n}_1 \\ &= (PDP^{-1})\mathbf{n}_1 \\ &= (PDP^{-1})(PDP^{-1})\mathbf{n}_0 \\ &= (PDP^{-1}PDP^{-1})\mathbf{n}_0\end{aligned}$$

Recalling that $P^{-1}P = I$, this last equation simplifies as

$$\begin{aligned}\mathbf{n}_2 &= PDIDP^{-1}\mathbf{n}_0 \\ &= PDDP^{-1}\mathbf{n}_0 \\ &= PD^2P^{-1}\mathbf{n}_0\end{aligned}$$

where D^2 is D multiplied with itself. Proceeding further, we see the general pattern:

$$(5) \quad \mathbf{n}_t = PD^tP^{-1}\mathbf{n}_0$$

Although Equation 5 looks more complicated than Equation 2, it is much easier to work with. In particular, as we saw in Example 8.4.3 and Exercise 8.4.7, for any diagonal matrix D , the matrix D^k is simply a diagonal matrix with each of the entries of D raised to the power of k . Therefore, once we have calculated the eigenvalues and eigenvectors of A , calculating the value of $\mathbf{n}_t$ is straightforward for any t of interest.

Not only does Equation 5 often provide a computational advantage over Equation 2, it can also be used to better understand the structure of the solution. Let's return to the case of $n = 2$. Writing Equation 5 more explicitly, we have

$$(6) \quad \mathbf{n}_t = \begin{bmatrix} | & | \\ \mathbf{v}_1 & \mathbf{v}_2 \\ | & | \end{bmatrix} \begin{bmatrix} \lambda_1^t & 0 \\ 0 & \lambda_2^t \end{bmatrix} P^{-1}\mathbf{n}_0$$

If we define two new constants, c_1 and c_2 , as

$$\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = P^{-1}\mathbf{n}_0$$

then Equation 6 can be written as

$$\begin{aligned}(7) \quad \mathbf{n}_t &= \begin{bmatrix} | & | \\ \mathbf{v}_1 & \mathbf{v}_2 \\ | & | \end{bmatrix} \begin{bmatrix} \lambda_1^t & 0 \\ 0 & \lambda_2^t \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \\ &= \begin{bmatrix} | & | \\ \mathbf{v}_1 & \mathbf{v}_2 \\ | & | \end{bmatrix} \begin{bmatrix} c_1\lambda_1^t \\ c_2\lambda_2^t \end{bmatrix} \\ &= c_1 \begin{bmatrix} | \\ \mathbf{v}_1 \\ | \end{bmatrix} \lambda_1^t + c_2 \begin{bmatrix} | \\ \mathbf{v}_2 \\ | \end{bmatrix} \lambda_2^t\end{aligned}$$